Tariffs Abroad, Transfers at Home
How Britain Absorbs a Trade Shock[*]

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Abstract

Who bears a foreign tariff once the tax–benefit system responds? I derive sector-level earnings shocks from the 2025 tariff schedule and US-export intensity by SIC division, and pass them through PolicyEngine UK on Family Resources Survey microdata. Holding the derived aggregate loss fixed, I vary the adjustment margin across four families: job loss, wage cuts, inactivity, and reallocation into services at a survey-calibrated 28.3 per cent earnings penalty. The system cushions wage cuts (41 per cent) more than earnings-equivalent job losses (33 per cent), reversing the usual ordering: income tax relieves wage cuts at marginal but job losses only at average rates, and Universal Credit reaches barely a third of displaced families. Nationally the shock is near-invisible—poverty rises 0.055 points—yet among affected households it runs from 4 to 43 per cent. A variant that discards the elasticity and lets the realised 2025–26 export falls be the shock halves the levels but leaves the margin ordering intact.

Keywords: tariffs, trade shocks, microsimulation, poverty, inequality, public finances

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1 Introduction

Who bears a foreign tariff shock after the tax–benefit system responds—and how much does the answer depend on whether workers adjust through job loss, wage cuts in place, exit into economic inactivity, or reallocation into lower-paid services work? In April 2025 the United States imposed a 10 per cent baseline tariff on UK goods, with 25 per cent rates on cars and steel; the May 2025 Economic Prosperity Deal (EPD) cut the auto rate to 10 per cent on a 100,000-vehicle quota, offered conditional steel relief, and—in a December 2025 side agreement paid for through NHS pricing concessions—exempted pharmaceuticals. A rich institutional literature has traced the macroeconomic, sectoral, regional and firm-level consequences of this shock (OBR, 2025; Bank of England, 2025; City-REDI, 2025; Centre for Cities, 2025). None of it reaches the household: no analysis passes the tariff through to disposable income, poverty, or the benefit system that would actually cushion it. This paper fills that cell.

The design is primitives-first. The shock enters as the tariff schedule itself, combined with US-export intensity by SIC division (ONS/HMRC trade-by-SIC) and an export-demand elasticity, to yield sector-level earnings shocks; the elasticity (2.0) is anchored on the realised outturn—US-bound goods exports fell 24.7 per cent in the first full tariff month against a 12.8 per cent trade-weighted tariff—and cross-checked against UK worker-level trade-shock estimates (De Lyon and Pessoa, 2021) and the OBR and Ignatenko et al. (2025) aggregate anchors. Employment is an *outcome* of the derived shock, not a free input. The derived sector shocks are then realised on Family Resources Survey workers in exposed industries under four adjustment-margin families: *displacement* into unemployment (the China-shock short run of Autor et al., 2013), *wage cuts with reallocation* into other sectors (the German and Danish long run of Dauth et al., 2021 and Traiberman, 2019), and *exit into inactivity* for older displaced workers—the margin UK deindustrialisation actually chose, when coalfield and steel job losses became benefit-financed economic inactivity rather than measured unemployment (Beatty et al., 2007; Beatty and Fothergill, 2017), and *reallocation into services*—the same displaced workers re-employed in retail, health, land transport and food service at an FRS-calibrated 28.3 per cent annual-earnings penalty, the margin the China-shock evidence actually describes (Autor et al., 2014; Dauth et al., 2021). The first three hold the aggregate earnings loss fixed; the fourth removes only the wage penalty on movers, and so is level-comparable with displacement (identical worker sets, paired draws) and rate-comparable with all. PolicyEngine UK, an open-source tax–benefit microsimulation model, computes disposable income, poverty, inequality and the Exchequer position under each.

The adjustment margin is the paper's central axis, and the results reverse the prior that motivated it. The naive reading of Universal Credit says the system should replace a job loss more generously than it tops up an in-work wage cut of equal aggregate size. On the microdata the opposite holds: the tax–benefit system cushions 41 per cent of the shock when it arrives as wage cuts but only 33 per cent when it arrives as displacement. An exact component decomposition demonstrates why: a wage cut is relieved at every worker's *marginal* income-tax rate (35 pence per pound of gross loss) while a job loss destroys tax liability only at the *average* rate (20 pence)—a gap that alone exceeds the headline difference—and Universal Credit, though it responds more to displacement than to wage cuts, reaches only a third of displaced workers' families, gated in the model by take-up and the joint partner-income taper (the capital means test cannot bind on FRS data, which carry no household capital). In the language of Atkin et al. (2025), statutory generosity toward the unemployed is not economic generosity once the gates bind on the population actually displaced. The marginal-versus-average asymmetry itself is not new—Auerbach and Feenberg (2000) originate the reading of the income tax as a marginal-rate stabiliser, McKay and Reis (2016) formalise the role of progressivity and of the intensive/extensive structure of shocks, and Dolls et al. (2012) find the tax term larger for the intensive shock in all nineteen EU countries and the United States. Three things here are: the *sign flip of the total*, as UK out-of-work support has shrunk enough that the always-present tax asymmetry now dominates and the ordering reverses; an exact, residual-free decomposition computed on the population *actually displaced* rather than on a proportional shock applied to everyone; and a gate-by-gate audit that counts which UC constraint binds on each displaced family. That audit also yields a methodological result with reach beyond this shock: our UK cushioning levels sit close to Dolls et al. (2012)'s UK cells, and the residual disagreement is almost entirely their full-take-up convention—forcing 100 per cent take-up in our own model recovers most of it—so full-entitlement microsimulation systematically overstates the stabilisation of the extensive margin, the only margin whose support has to be claimed. Two scope limits travel with the design: the wage-cut family models the earnings consequence of reallocation without moving anyone between sectors, and household behavioural margins—added-worker responses, retirement timing, self-employment entry (Irastorza-Fadrique et al., 2025)—are suppressed throughout. Whichever margin, the calibrated shock (a £1.8 billion annual gross earnings loss, 0.06 per cent of GDP on the direct channel) reaches households mainly as a fiscal event—£0.7–1.0 billion a year to the Exchequer (Monte Carlo means over 50 displacement draws)—rather than a poverty event (at most +0.06 percentage points); and on the displacement margin it is regionally, not vertically, concentrated, with losses landing in deciles 7–10 of the income distribution while averaging £60–110 per person per year in the hardest-hit regions—and reaching £100–200 per head in the few regions any single realisation lands on. Which margin the data chose is testable against the 2025–26 outturn (ONS trade, BICS, SMMT monthly series), and the historical UK evidence decomposing displacement costs into wage cuts versus non-employment spells (Rud et al., 2022) calibrates the split.

Two sectoral nuances discipline the headline claims. Steel carries a caveat: roughly 80 per cent of UK steel exports go to the EU, so the steel–Wales cell is more exposed to EU/UK safeguards than to US tariffs. And pharmaceuticals—exempted under the December 2025 deal in exchange for VPAG rebate cuts and NHS pricing concessions—is a fiscal and pricing channel, not an employment channel, and is treated as such. Finally, an EPD counterfactual (full tariffs versus deal-mitigated rates) prices what the deal was worth, sector by sector and region by region, to households and the Treasury. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment signs; we compute first-round fiscal incidence holding fiscal policy fixed.

The remainder of the paper proceeds as follows. Section 2 situates the study between the quantitative trade-incidence literature and tax–benefit microsimulation. Section 3 sets out the tariff schedule, the trade-by-SIC exposure mapping, the derived sector shocks and the four adjustment-margin families. Section 4 presents the results: the derived shocks and their outturn anchor, the cushioning reversal, the distributional, fiscal and regional incidence by margin, and the EPD counterfactual. Section 5 turns to the EPD counterfactual and policy responses. Section 6 concludes.

2 Literature

This paper sits at the intersection of four literatures: the quantitative and reduced-form work on the incidence of trade shocks; the evidence on how workers actually adjust to them; the institutional analyses of the 2025 US tariffs on the UK; and the microsimulation literature on tax–benefit cushioning of earnings shocks. The intersection itself is empty: no study passes a trade-policy shock through a full tax–benefit microsimulation to household disposable income, poverty and the Exchequer.

2.1 Trade shocks and their incidence

Two recent papers anchor the structural end and frame the adjustment-margin question as its poles. Ignatenko et al. (2025) build a quantitative general-equilibrium model—nesting Armington, Eaton–Kortum, Krugman and Melitz structures with exact hat algebra over 194 countries—in which the 2025 tariffs enter as a *primitive* on trade shares. Wages are flexible and labour supply elastic, so employment is an equilibrium *outcome*; strikingly, its sign flips with revenue recycling (a tariff-financed income-tax cut raises US employment by 0.32 per cent, a lump-sum rebate lowers it by 0.41 per cent), and the optimal US tariff is a uniform 19 per cent independent of bilateral deficits. The UK is in their sample, and their replication data provide the UK-specific welfare and employment cells we use as aggregate anchors. Atkin et al. (2025), by contrast, apply a Baqaee–Farhi wedge framework to Chilean administrative data with factor supplies *fixed*: there is no employment margin at all, all adjustment runs through flexible wages and prices, and household incidence arrives via consumption baskets, factor ownership, profits and transfers—their "statutory versus economic incidence" framing is the one we adopt. One paper derives employment from the tariff primitive; the other assumes it fixed and lets wages absorb everything. A third structural anchor sits between the poles: Rodríguez-Clare et al. (2025) embed the 2025 trade war in a dynamic trade-and-reallocation model with *downward nominal wage rigidity*, in which higher tariffs raise US manufacturing employment but lower total employment as falling real wages depress participation, with US real income down about 1 per cent by 2028—rigid wages turn the same primitive into an employment shock that flexible-wage models turn into a wage shock, which is precisely the disagreement our adjustment-margin axis makes an explicit scenario dimension. On the partner-country side, Michelena et al. (2025) run the 2025 escalation through a multiregional input–output framework and find employment and export losses concentrated in the *targeted* countries—the same statutory-versus-economic distinction at the country level, but at the level of sectoral aggregates rather than households. Complementing the model-based work, den Besten and Känzig (2025) construct a narrative series of US tariff shocks over 1840–2024 and find tariff increases robustly contractionary—imports fall sharply, exports decline with a lag, and output drops persistently—historical discipline for treating the shock as a negative export-demand shock rather than a benign relative-price change.

The reduced-form lineage begins with Autor et al. (2013): import competition caused large, persistent employment and wage losses in exposed US commuting zones, with effects persisting to 2019 and no offsetting migration (Autor et al., 2021). Caliendo et al. (2019) provide the structural counterpart—a dynamic spatial model with costly mobility in which the China shock is a productivity/trade-cost primitive, generating aggregate gains alongside concentrated regional losses—and Adão et al. (2023) supply the theory-consistent mapping from regional shift-share estimates to aggregate counterfactuals that cautions against reading regional results as national ones. Kim and Vogel (2021) translate reduced-form employment and wage responses into money-metric worker welfare, finding regressive incidence with displaced workers' losses exceeding wage losses; Galle et al. (2023) show in a Roy-model setting that aggregate gains coexist with group-specific losses. Our contribution relative to this structural incidence work is to replace stylised worker groups and stylised transfers with real households and the actual UK tax–benefit system.

The 2018–19 US tariff episode supplies the sharpest direct evidence on who pays a tariff. Pass-through into US import prices was essentially complete, placing the incidence on the importing country's firms and consumers (Amiti et al., 2019; Fajgelbaum et al., 2020), with retail prices adjusting more slowly than border prices (Cavallo et al., 2021)—a pattern the 2025 wave repeats: matching daily retailer microdata to product-level tariff rates, Cavallo et al. (2025) estimate retail pass-through of about 20 per cent by September 2025, contributing roughly 0.7 percentage points to the US CPI; and Flaaen and Pierce (2019) show that US manufacturing employment fell on net, as input-cost increases and foreign retaliation outweighed import protection. The retaliation side of that episode is the closest analogue to our channel—a partner's tariff arriving as an export-demand shock: Waugh (2019) finds Chinese retaliatory tariffs depressed consumption in exposed US counties, and Blanchard et al. (2019) trace the electoral fallout of retaliatory tariffs across US counties. The broader distributional literature—surveyed in Fajgelbaum and Khandelwal (2016) and Fajgelbaum and Khandelwal (2022), with Borusyak and Jaravel (2021) decomposing expenditure against earnings channels—establishes that trade shocks redistribute along both consumption and income margins. All of this work prices tariffs at the border, the store, or the county; none of it passes the shock through a household-level tax–benefit system, which is the step taken here. Two propagation channels our direct-channel design deliberately excludes have well-measured analogues in this literature. Firm-level shocks travel along supply chains—Barrot and Sauvagnat (2016) show idiosyncratic disruptions imposing substantial output losses on customers of affected suppliers, and Acemoglu et al. (2016) find network effects on aggregate outcomes several times larger than the direct effect—and tradable-sector job losses multiply into local services, with each lost tradable job costing roughly 0.4–1.6 additional local jobs across the US and German estimates (Moretti, 2010; Gathmann et al., 2020). Both channels scale our direct-channel numbers upward; Section 6 carries the caveat. Open-economy macro adds an offsetting margin: exchange rates absorb part of a tariff, but imperfectly and state-dependently—at most a fifth of the 2018–19 dollar appreciation is attributable to the tariffs (Jeanne and Son, 2024), the dollar appreciates on unilateral tariffs but depreciates under retaliation (Corsetti et al., 2025), and tariff *uncertainty* itself depreciated the dollar in 2025 through a risk-premium channel (Kalemli-Özcan et al., 2026); model-based analyses of retaliation similarly find non-trivial consumption losses for passive trade partners (Auray et al., 2020). We hold the exchange rate fixed and let the outturn-anchored elasticity absorb whatever FX offset in fact occurred. Early assessments of the 2025 wave itself (Fajgelbaum and Khandelwal, 2026) again find the incidence landing on the tariff-imposing country's consumers—complementary to our question, which is the incidence on the tariffed *partner's* workers.

2.2 Adjustment margins: how workers absorb trade shocks

The worker-level evidence says the margin is mostly earnings, not permanent unemployment. Autor et al. (2014) find US workers exposed to import competition lose earnings amid churn across employers and sectors rather than through indefinite joblessness. Traiberman (2019) shows for Denmark that occupation-specific human capital produces reallocation with persistent wage losses; because our shock is an *export-demand* shock rather than import competition, the most direct microeconometric precedents are Hummels et al. (2014), who instrument Danish firm-level export demand and trace it into worker wages by task, and Garin and Silvério (2024), who show that Portuguese exporters pass idiosyncratic export-demand shocks into the wages of incumbent workers rather than shedding them—firm-level evidence for exactly the wage-cut family of our scenario grid. Dauth et al. (2021) provide the cleanest evidence for the wage-margin family, with German workers reallocating into services and other plants at lower wages. Dix-Carneiro (2014) estimates that reallocation takes years, with adjustment costs of 1.4–2.7 times annual wages and older workers hit hardest—calibrating our transition horizons and age gradients—while Dix-Carneiro and Kovak (2017) show regional effects that *grow* over two decades on a joint wage/employment margin.

The UK's own history points to a third margin absent from the US and German studies: benefit-financed economic inactivity. Beatty et al. (2007) document that coalfield job losses became male inactivity on incapacity benefits, not measured unemployment, with incomplete recovery after twenty years; Beatty and Fothergill (2017) show that 1980s–90s industrial job destruction *still* inflates the deficit through higher benefit claims and lower tax receipts—the direct UK precedent for our fiscal-incidence question, retrospective and area-level where ours is prospective, rules-based and household-level. The margin is not merely historical: Roberts and Taylor (2022) show in 2009–18 longitudinal data that disability benefit claims still respond to local labour-market slack over and above health status, Beatty and Fothergill (2023) document the persistence of hidden unemployment among incapacity claimants across large parts of Britain, and the IFS records 4.2 million working-age people on health-related benefits by 2024, up from 3.2 million in 2019 (IFS, 2024)—the inactivity margin is live at exactly the moment the tariff shock arrives. Aragón et al. (2018) add gender-differentiated margins from mine closures, and Rud et al. (2022) decompose displacement costs into wage cuts on re-employment versus non-employment spells—directly calibrating the split between our displacement and wage-cut families. For UK-specific elasticities, De Lyon and Pessoa (2021) is the key source: UK workers in Chinese-import-competing industries lost employment years and earnings without equally gainful re-employment, with larger female losses through employment years (see also Dorn and Levell, 2024). Closest to our object, Irastorza-Fadrique et al. (2025) study UK *household* responses to trade shocks—added-worker effects, later retirement, self-employment—but model behaviour, not tax–benefit mechanics; we treat their margins as complementary to our rules-based incidence.

2.3 The 2025 tariffs and the UK

The institutional literature on the 2025 shock is rich but stops above the household. NIESR's NiGEM scenarios and the OBR's March 2025 tariff analysis provide the macro anchors—the OBR puts the permanent UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent (NIESR, 2024; OBR, 2025). The Bank of England estimates the effective US tariff on the UK at roughly 9 per cent (against about 18 per cent globally) and finds trade diversion has mildly *lowered* UK import prices (Bank of England, 2025; Dhingra, 2025)—which is what licenses our isolating the export-demand/earnings channel while holding the price and monetary channels fixed; the Bank's own staff work on the exchange-rate response to tariffs and retaliations (Corsetti et al., 2025) supplies the FX side of that held-fixed channel. The multilateral institutions frame the aggregates: the IMF's October 2025 outlook has global growth slowing to 3.2 per cent in 2025 amid the tariff shock (IMF, 2025), and the WTO's October 2025 update puts merchandise trade volume growth at 2.4 per cent in 2025 but only 0.5 per cent in 2026 as the tariffs bind (WTO, 2025). Firm surveys find most UK firms expecting small negative effects, with the US taking about 22 per cent of UK exports (CEP, 2025). The regional and sectoral geography is sharp: Coventry sends 22 per cent of its exports to the US (Centre for Cities, 2025); auto tariffs are estimated to cost £10bn of GDP over five years with the West Midlands bearing 62 per cent of it

(City-REDI, 2025); SMMT data show US vehicle exports collapsing from 7,077 units in April to 4,223 in May 2025 before the EPD's implementation (SMMT, 2025). UK Steel notes that some 80 per cent of UK steel exports go to Europe—the basis of our steel caveat—and the December 2025 pharmaceutical deal (0 per cent tariffs for three years, paid for through VPAG rebate cuts from 22.5 to 14.5 per cent and NHS pricing concessions) makes pharma a fiscal/pricing shock rather than an employment shock (ABPI, 2025). The Resolution Foundation's household-facing analysis is descriptive rather than microsimulated (Resolution Foundation, 2025a); the IFS has no dedicated distributional publication on the 2025 tariffs. Across macro, sector, region and firm, the household/fiscal cell is empty.

For UK trade-shock precedents more broadly, Dhingra et al. (2017) map Brexit trade costs to income losses, and Fetzer and Wang (2024) provide the best district-level trade-shock incidence estimates (synthetic-control output losses of 5–10 points concentrated in manufacturing districts); Fetzer (2019) and Bloom et al. (2019) supply the political-economy and uncertainty channels.

2.4 Tax–benefit cushioning of earnings shocks

The methodological ancestors are Dolls et al. (2012), who use EUROMOD and TAXSIM to show EU tax–benefit systems absorb on average about 38 per cent of a proportional income shock and 47 per cent of an unemployment shock (against roughly 32 per cent in the US). The EU averages are not the right comparator for this paper; their *UK* cells are, and they are close to ours: the UK income-shock stabilisation coefficient is 0.352 (income tax 0.267, social insurance contributions 0.054, benefits 0.031) and the UK unemployment-shock coefficient 0.415 (tax 0.191, SIC 0.061, out-of-work benefits 0.163). Our 41.0 per cent (wage cut) and 32.8 per cent (displacement) sit alongside those levels; where we disagree is the *ordering*, and Section 4.2 traces that disagreement to a single component. The interpretation of the tax term as an effective *marginal* rate is theirs too—and older still: Auerbach and Feenberg (2000) originate the framing of the income tax as an automatic stabiliser operating at marginal rates, and McKay and Reis (2016) formalise how progressivity and the extensive/intensive structure of shocks determine stabilisation in general equilibrium. The asymmetry between marginal and average relief is therefore not our discovery; what is new here is that in the UK of the 2020s it is large enough to *flip the sign* of the intensive–extensive ordering. A second ancestor is Sutherland and Figari (2013) on EUROMOD's stress-test use—but both apply stylised macro shocks, not trade-policy-derived ones. Our direct methodological sibling is Brewer and Tasseva (2021), who pass the Covid-19 labour-market shock through UKMOD to show that the UK tax–benefit system plus the emergency support schemes held the rise in income inequality far below the rise in earnings inequality: the same architecture—an externally derived shock realised on survey households and run through the full UK benefit rules—applied to a pandemic where ours applies it to a trade-policy shock, without the emergency schemes and with the adjustment margin as the central axis. The wider macro-micro linkage literature supplies the lineage for the shock-derivation step itself: Bourguignon et al. (2008) codify the techniques for passing macro shocks to household microdata, Peichl (2016) surveys the linkage of microsimulation to structural macro models, and Navicke et al. (2013) show how EUROMOD nowcasts impose estimated labour-market transitions on survey microdata—the same device as our displacement draws, applied there to observed rather than counterfactual shocks. On the US side, exposed commuting zones saw rising disability, retirement and welfare payments that offset only a small fraction of earnings losses (Autor et al., 2013); Hyman et al. (2024) show wage insurance raises re-employment and earnings, a natural counterfactual reform for our framework. The nearest tariff-microsimulation links are on the US side: the Tax Policy Center's tariff models allocate tariff burdens to tax units through consumption baskets in a full microsimulation model (Tax Policy Center, 2025), the Budget Lab at Yale prices the 2025 schedule's fiscal and distributional effects by income decile (Budget Lab, 2025), and Acosta and Cox (2024) document that the US tariff code is built regressively, with higher rates on lower-end varieties of the same goods. Two institutional facts underpin the cushioning asymmetry our results turn on. UC's capital rules—thresholds frozen in cash terms since 2006—reduced or extinguished entitlement for around two million income-eligible families, with roughly 1.1 million standing to gain from uprating alone (Resolution Foundation, 2025b); and take-up of income-related benefits is well below 100 per cent (DWP, 2024), so statutory replacement rates overstate what displaced workers actually receive. Both push in the direction of our finding that the means-tested extensive margin cushions less than the automatic in-work margin—take-up gates entitlement inside the model, while the capital rules are a real-world gate the plain FRS data cannot exercise (Section 6). But all of these allocate the *consumer-price* incidence of a country's own tariffs; ours is the export-demand/earnings incidence of a *partner's* tariff. The distinction between these channels, and the absence of any analysis in ours, defines the gap this paper fills: structural incidence models have no realistic fiscal system, microsimulation has no trade shock, and the adjustment-margin axis has not been quantified by anyone. That axis is where this paper's prior gets tested. The natural reading of the two institutional facts above, and of Dolls et al. (2012)'s own unemployment-versus-income ordering, is that the system should replace a job loss more generously than an earnings-equivalent in-work wage cut, because Universal Credit exists for the former and only tapers against the latter. Section 4.2 shows that on UK microdata the ordering runs the other way, and Section 4.3 shows why.

3 Methodology

3.1 Design: primitives first

A reader is entitled to ask, before any number is reported, which parts of this design are read off data and which are assumed. The answer, stated in full:

Object	Status
Tariff schedule τ_j	**Data** (policy documents: the April 2025 schedule, the May 2025 EPD, the December 2025 pharmaceutical agreement).
US-export intensity x_j	**Data** (HMRC RTS/OTS customs exports over ONS ABS division turnover; Appendix A).
Export fall	**Data** in the measured family (realised HMRC outturn, May 2025–February 2026); **calibrated** in the tariff families, via an elasticity anchored on the April 2025 outturn.
Pass-through π	**Assumed** ($\pi = 1.0$), varied on a grid (Appendix D).
Adjustment margin	**Assumed**, and deliberately varied. This is the paper's free dimension.
Household incidence	**Computed** from statutory tax and benefit rules in PolicyEngine UK; no behavioural or reduced-form step.

The margin is assumed because the literature does not agree on it, and the disagreement is structural rather than empirical: Ignatenko et al. (2025) derive the employment response from elastic labour supply and flexible wages, while Rodríguez-Clare et al. (2025) derive it from downward nominal wage rigidity, so the same tariff primitive arrives as a wage shock in the first model and a job shock in the second. We do not endogenise the margin. We price the disagreement—holding the aggregate earnings loss fixed and reporting what each pole implies for households and the Exchequer. That is a weaker claim than a structural model makes, and a more auditable one.

The shock enters through primitives, not through an assumed employment effect. The pipeline is:

$$s_j \;=\; \tau_j \, x_j \, \varepsilon \, \pi, \tag{1}$$

where s_j is the derived earnings-shock size for SIC 2007 division j (the fraction of the division's employee wage bill removed by the tariff), τ_j is the tariff rate faced by the division under the schedule, x_j is its US-export intensity, ε is the export-demand elasticity, and π is the pass-through of the sector output loss to the sector wage bill. x_j is US goods exports of the division as a share of division turnover, built from the HMRC uktradeinfo RTS/OTS API (2024 US exports, £52.5 billion mapped to SIC divisions of a £55.6 billion customs-basis total, via a documented SITC-to-SIC crosswalk) and ONS Annual Business Survey turnover denominators; Appendix A gives the full provenance and validation. The elasticity $\varepsilon = 2.0$ is anchored on the realised outturn: ONS data show UK goods exports to the US falling 24.7 per cent in April 2025, the first full tariff month, against a 12.8 per cent trade-weighted average tariff implied by the schedule and the built intensities ($0.247/0.128 = 1.93$, rounded to 2.0, mid-range of short-run trade elasticities). The pass-through $\pi = 1.0$ places the full sector output loss on the sector wage bill—the appropriate short-run displacement assumption. Because ε and π enter Equation (1) only through their product, all pound-denominated results scale approximately linearly in $\varepsilon\pi$, while cushioning *rates* and margin orderings are essentially invariant to it; Appendix D verifies both properties on a 4×3 sensitivity grid over $\varepsilon \in \{1.0, 1.5, 2.0, 3.0\}$ and $\pi \in \{0.5, 0.75, 1.0\}$. Since the central $\varepsilon = 2.0$ is anchored on a single month's realised outturn (April 2025), the grid doubles as the uncertainty band around that anchor. The implied aggregate gross earnings loss under the full schedule is £1.77 billion a year, about 0.06 per cent of GDP on the direct exposed-sector channel, sensibly below the OBR's all-channel scenario range of one-third to three-quarters of a per cent (rising to about 1 per cent at peak in its reciprocal-tariff scenario) (OBR, 2025) and consistent with the UK cell of Ignatenko et al. (2025) and UK worker-level responses (De Lyon and Pessoa, 2021; Rud et al., 2022). Employment is then an *outcome* of s_j under an assumed adjustment margin—never a free input. The design covers the direct exposed-sector channel only: local-multiplier estimates of 0.4–1.6 additional local jobs per lost tradable job (Moretti, 2010; Gathmann et al., 2020) imply the all-channel employment effect is larger, so the derived shock is a lower bound (Section 6). Appendix F bounds the first omitted channel directly, propagating the export-demand falls upstream through the ONS domestic input-output tables via a Leontief round (Barrot and Sauvagnat, 2016; Acemoglu et al., 2016).

3.2 The tariff schedule and the EPD counterfactual

Two tariff scenarios bracket the policy. *Full tariffs*: the April 2025 schedule—10 per cent baseline on UK goods, 25 per cent on motor vehicles (SIC 29) and steel (SIC 24). *EPD*: the deal-mitigated schedule—autos at the 10 per cent in-quota rate (100,000 vehicles, roughly the 2024 US export volume, so the quota binds only marginally); steel with conditional, partial relief; pharmaceuticals (SIC 21) exempt under the December 2025 agreement. The difference between the two scenarios, run through identical downstream machinery, prices the EPD for households and the Exchequer. Two caveats travel with the schedule. Steel: about 80 per cent of UK steel exports go to the EU, so the US tariff is a minority driver of the sector's exposure and the steel cell should be read alongside EU/UK safeguards. Pharmaceuticals: the exemption was purchased through VPAG rebate cuts and NHS pricing concessions, so pharma's incidence is fiscal and price-side, not an employment shock; we model its employment shock as zero under the EPD and discuss the fiscal channel separately (Section 5).

3.3 Adjustment-margin families

The derived s_j is realised on Family Resources Survey 2024–25 employees in exposed divisions under four families, the first three of which each hold the aggregate earnings loss $\sum_j s_j B_j$ (with B_j the division wage bill) fixed:

Displacement (ADH short run). Within each exposed division a weighted head-count quota $s_j \times$ (division employee weight) is drawn with uniform ordering keys, so a represented worker's displacement probability does not depend on their record's grossing weight; displaced workers move to unemployment with earnings, hours, pension contributions and statutory pay zeroed.

Wage cuts with reallocation (Dauth/Traiberman long run). No job loss; every employee in division j takes a proportional cut of s_j, so the weighted aggregate earnings removed equals $\sum_j s_j B_j$ by construction—exact earnings-equivalence with the displacement family's expected removal.

Inactivity (Beatty–Fothergill UK history). The displacement draw, but displaced workers aged 50 and over exit to economic inactivity, the benefit-financed route of UK deindustrialisation, rather than unemployment. The route is benefit-financed in the model, not just relabelled: every inactive worker is flagged as having limited capability for work-related activity, so their household's Universal Credit includes the LCWRA health element (£217.26 a month in 2026; in policyengine-uk 2.89.2 the stored survey-imputed `uc_LCWRA_element` is overridden with the baseline element plus one annual LCWRA amount per benefit unit containing any newly flagged member—at most one element per benefit unit, as the rules provide—leaving all other households' stored elements untouched, and the run hard-errors if the element fails to reach any flagged person's benefit unit). This mirrors the Beatty et al. (2007) route by which industrial job loss became incapacity-benefit inactivity. The strong assumption should be stated plainly: *all* displaced workers aged 50 and over are assumed to pass the Work Capability Assessment, so the family is an upper bound on the health-element route; a sensitivity variant in which only 50 per cent of the older displaced qualify is reported alongside (Section 4.4). Work-search conditionality carries no separate fiscal machinery in the model, so the LCWRA element is the entire modelled wedge between inactivity and unemployment.

Reallocation into services (ADH/Autor et al., 2014). The same displacement draw on the same seed—so the two families are paired worker-for-worker—but the drawn workers are re-employed rather than left out of work. Their `sic_industry_division` is reassigned to one of four services destinations (SIC 47 retail, 86 human health, 49 land transport, 56 food and beverage service) in proportion to those divisions' FRS employee-weight shares (29.5, 45.8, 9.2 and 15.5 per cent), and their annual earnings are cut by a penalty calibrated on the FRS itself: weighted mean annual employee earnings of £48,272 over 2,047 hours in the exposed goods divisions against £34,594 over 1,701 hours in the four destinations, a **28.3 per cent annual-earnings penalty** that embeds the hours fall. Two variants are reported: an hours- and age-controlled **low-penalty** case of 14.0 per cent (the pure hourly-wage gap), and a **three-month lag** in which workers are out of work for a quarter of the year before the services job starts, scaling annual earnings and hours by a further $1 - 3/12$. ASHE 2024 full-time medians bracket the hours-controlled figure at 10–20 per cent; the FRS all-employee penalty is larger because it also captures the shift into part-time work. Because this family removes only the penalty on the movers rather than the full earnings-equivalent aggregate, its pound figures are comparable with the displacement family (identical worker sets, paired draws) but *not* with the wage-cut family; its cushioning *rate* is comparable with both. Section 4.9 reports the results and Appendix B the mechanics.

Two scope limits on the family set should be stated here rather than in a footnote. The wage-cut family models the *earnings consequence* of reallocation without moving anyone between sectors: workers keep their observed industry and hours, and the proportional cut stands in for the wage penalty that a completed reallocation would carry; the reallocation family does move them, but who reallocates is assumed (it is the displacement draw) rather than estimated. And no household-side behavioural margin is active—added-worker responses, earlier or later retirement, and shifts into self-employment (Irastorza-Fadrique et al., 2025) are all suppressed, as are any labour-supply responses to the tax and benefit changes themselves. All four families are therefore first-round, rules-based incidence.

The margin is the paper's central axis because the tax–benefit system treats the two poles asymmetrically: an in-work earnings reduction is offset automatically at marginal tax and taper rates, while a job loss moves the worker to the gated, means-tested extensive margin of the benefit system. Which pole the system cushions more is an empirical question the microdata answer—in the direction opposite to the naive replacement-rate intuition (Section 4.2).

3.4 Microsimulation and outcomes

The realisation step—an externally derived shock imposed as status and earnings transitions on survey households, then run through the full benefit rules—follows Brewer and Tasseva (2021), who apply it to the Covid-19 labour-market shock in UKMOD, and the EUROMOD nowcasting tradition of imposed labour-market transitions (Navicke et al., 2013). PolicyEngine UK computes baseline and shocked household disposable income (HBAI cash concept throughout), relative before-housing-costs poverty, absolute before-housing-costs poverty against a fixed baseline line (computed manually, since the installed model version exposes no absolute-poverty variable; Appendix C), after-housing-costs poverty, the Gini coefficient of equivalised disposable income, decile and region breakdowns, and the Exchequer effect (foregone tax plus additional benefit spending, via the government balance). Displacement draws are Monte Carlo: every displacement/inactivity result is reported as the mean and standard deviation over n draws (default 20). Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment responses; we compute first-round incidence holding fiscal policy fixed and treat recycling as a policy counterfactual. One scope limitation on the fiscal aggregate should be stated: only direct taxes and benefits respond to the shock. Indirect taxes (VAT and duties) are not shocked, so the consumption-tax revenue foregone on the households' reduced spending is absent from the government balance and the reported Exchequer cost is an understatement of the true first-round fiscal effect.

4 Results

All results are computed on Family Resources Survey 2024–25 microdata for the 2026 period, on the HBAI cash disposable income concept throughout (Section 3.4). Displacement results are means over 50 Monte Carlo draws (inactivity results over 20 draws, seeds 0–19) with ± values denoting standard deviations across draws; the wage-cut margin is deterministic. Decile and regional profiles are Monte Carlo means across the 50 displacement draws. Because the draw-to-draw standard deviation on the Exchequer cost is large (about £0.30 billion against means of £0.7–0.9 billion), I narrate only orderings and contrasts that survive it, and flag those that do not.

I lead with the paper's headline finding, because it reverses the prior that motivated the design. Section 3.3 argued—as the naive reading of Universal Credit suggests—that the tax–benefit system should replace a job loss more generously than an in-work wage cut of equal aggregate size. On the FRS microdata the opposite holds: the system cushions 33 per cent of the gross earnings loss when the shock arrives as displacement, but 41 per cent when the identical aggregate loss arrives as proportional wage cuts. Section 4.3 decomposes the gap component by component. Two mechanisms carry it. First, tax arithmetic: a wage cut is relieved at every affected worker's *marginal* income-tax rate (35 pence per pound of gross loss), whereas a job loss wipes out the whole earnings stream—personal allowance and all—so the income-tax offset falls to the *average* rate (20 pence). Second, the extensive margin of the benefit system gates out-of-work support: two-thirds of displaced workers' families receive no UC at all post-shock, so UC restores only 4 pence per pound of displacement loss—though the binding gates in the model are take-up and the joint partner-income taper, not the capital test, which the FRS data cannot exercise. The adjustment margin therefore determines not just *where* the shock lands but *how much of it* the state absorbs—and in the direction opposite to the standard intuition.

4.1 The derived sector shocks

Table 1 reports the primitives and the derived shock for the most exposed SIC 2007 divisions. US-export intensity—2024 US goods exports over division turnover, built from the HMRC RTS/OTS API and the ONS Annual Business Survey as described in Appendix A—is highest in machinery (0.180), computer/electronic and optical products (0.179), other transport equipment including aerospace (0.157), pharmaceuticals (0.130), electrical equipment (0.126), chemicals (0.125) and motor vehicles (0.110); basic metals including steel, despite its 25 per cent tariff, has intensity of only 0.051 because most UK steel output does not go to the US. With the calibrated elasticity $\varepsilon = 2.0$ and full wage-bill pass-through (Section 3.1), the derived earnings shocks $s_j = \varepsilon \tau_j x_j$ under the full schedule are led by autos (5.5 per cent of the division wage bill), machinery (3.6 per cent), electronics (3.6 per cent), aerospace/other transport (3.1 per cent), pharmaceuticals (2.6 per cent) and steel (2.5 per cent). Under the EPD, the auto shock falls to 2.2 per cent, steel to 1.3 per cent, and pharmaceuticals to zero.

The anchor check. Applied to FRS 2024–25 employees, the full schedule removes £1.77 billion of gross annual earnings—about 0.06 per cent of GDP. This is the *direct exposed-sector earnings channel only*. Under the EPD schedule the gross earnings loss is £1.39 billion.

It is worth putting the paper's three aggregates on one line, as shares of GDP, against the macro benchmark. The direct exposed-sector earnings channel is 0.06 per cent (full schedule);

Table 1: Primitives and derived earnings shocks, most exposed SIC divisions

SIC	Division	Tariff τ_j Full	Tariff τ_j EPD	US-export intensity x_j	Derived shock s_j (%) Full	Derived shock s_j (%) EPD
29	Motor vehicles	0.25	0.10	0.110	5.5	2.2
28	Machinery and equipment	0.10	0.10	0.180	3.6	3.6
26	Computer, electronic, optical	0.10	0.10	0.179	3.6	3.6
30	Other transport (incl. aerospace)	0.10	0.10	0.157	3.1	3.1
21	Pharmaceuticals	0.10	0.00	0.130	2.6	0.0
27	Electrical equipment	0.10	0.10	0.126	2.5	2.5
20	Chemicals	0.10	0.10	0.125	2.5	2.5
24	Basic metals (incl. steel)	0.25	0.125	0.051	2.5	1.3
11	Beverages	0.10	0.10	0.063	1.3	1.3

Notes: $s_j = \varepsilon\, \tau_j\, x_j$ with $\varepsilon = 2.0$ and pass-through 1.0. Intensity is 2024 US goods exports over ONS ABS division turnover (Appendix A). The EPD steel rate is the parameterised half-relief of the 25 per cent rate; the auto rate is the in-quota 10 per cent.

the measured-outturn family, which imposes no elasticity at all, is £0.95 billion, about 0.03 per cent (Section 4.11); and the supply-chain-amplified shock, adding one upstream Leontief round, is £3.45 billion, about 0.12 per cent (Appendix F). The OBR's March 2025 analysis puts the *permanent* UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent, reaching roughly 1 per cent at peak in its reciprocal-tariff scenario (OBR, 2025). Our figures sit below that range, as they should: the OBR's is an all-channel general-equilibrium estimate including uncertainty, investment and trade-diversion channels a static microsimulation deliberately excludes. The gap is smaller than it looks, however, because the omitted channels are broadly *additive* rather than alternative: the upstream Leontief round is a factor of 1.95 on the direct channel, and local-multiplier estimates of 0.4–1.6 additional local jobs per lost tradable job (Moretti, 2010; Gathmann et al., 2020) are a further factor of roughly 1.4–2.6 on top of it. The two factors compound without double-counting because the 1.95 is Type-I-equivalent—direct plus indirect upstream requirements, with no induced-consumption round—so the induced channel enters only once, through the local multipliers. For scale, ONS publishes Type I output multipliers for manufacturing of roughly 1.4–2.0, and 1.95 sits at the top of that range, as one would expect for the input-intensive exposed manufacturing divisions rather than manufacturing as a whole. Compounding accordingly gives an all-channel figure of about 2.7–5.1 times the direct earnings shock—0.16 to 0.31 per cent of GDP—approaches the OBR range from below, with investment, uncertainty and general-equilibrium price channels still to add. The elasticity itself is anchored on the realised outturn rather than borrowed from the literature: UK goods exports to the US fell 24.7 per cent in April 2025 against a trade-weighted average full-schedule tariff of 12.8 per cent computed from Table 1's primitives, giving $0.247/0.128 = 1.93$, rounded to 2.0—also mid-range of short-run trade elasticities (Section 4.10). Because the elasticity and the pass-through enter the derived shock only through their product, the pound levels above scale approximately linearly in $\varepsilon\pi$ while every rate and ordering in this section is essentially invariant to it: Appendix D verifies this on a 4×3 grid over $\varepsilon \in \{1.0, 1.5, 2.0, 3.0\}$ and $\pi \in \{0.5, 0.75, 1.0\}$, which also serves as the uncertainty band around the one-month outturn anchor.

Realised as displacement, the full schedule puts 34,200 workers (±3,600 across 50 draws) out of work; the EPD schedule 27,700 (±4,800; the quota expectations are 33,566 and 26,709). The age profile of the displaced skews old relative to the workforce—across draws, 18 ± 10 per cent are aged 55–64 and 23 ± 12 per cent are 55 or over—which is what gives the inactivity margin its bite: a third of the displaced are 50 or over and flow to benefit-financed inactivity with the UC health element (Section 4.4).

4.2 Cushioning: the tax–benefit system absorbs wage cuts more than job loss

Table 2 makes the headline finding precise. Define the cushioning rate as one minus the ratio of the aggregate household disposable income loss to the aggregate gross earnings loss. Under the full schedule, the displacement margin turns a £1.70 billion gross earnings loss (seed-0 draw) into a £1.14 billion disposable income loss—a cushioning rate of 32.8 per cent—while the earnings-equivalent wage-cut margin turns £1.77 billion of gross earnings into a £1.04 billion disposable loss, a cushioning rate of 41.0 per cent. The same ordering holds under the EPD schedule (35.1 versus 40.8 per cent), so it is robust to the composition of the shock across sectors.

Table 2: Cushioning rates by tariff scenario and adjustment margin (seed 0)

Tariff scenario	Margin	Gross earnings loss (£bn/yr)	Net disposable loss (£bn/yr)	Cushioning rate (%)
Full tariffs	Displacement	1.70	1.14	32.8
Full tariffs	Wage cut	1.77	1.04	41.0
EPD	Displacement	0.91	0.59	35.1
EPD	Wage cut	1.39	0.82	40.8

Notes: cushioning rate = 1 − (net disposable loss)/(gross earnings loss), aggregated over all households. Displacement rows are the seed-0 draw (the wage-cut margin is deterministic); the displacement gross loss differs slightly from the expected £1.77bn/£1.39bn because a single draw realises the quota on discrete workers. The EPD displacement gross loss is smaller than the EPD wage-cut loss for the same reason compounded by which sectors' quotas round down at seed 0.

This result reverses the prior stated in Section 3.3, and rather than assert a mechanism I demonstrate it: Section 4.3 decomposes both cushioning rates into named tax and benefit components, counts which UC constraint binds on each displaced worker's family, and reruns the displacement scenario with the UC capital rules switched off. To preview the conclusion: the gap is carried by income-tax arithmetic (marginal versus average rates) reinforced by the gated extensive margin of UC—where the gates that actually bind in the model are modelled take-up and the joint partner-income taper, not the capital test. In the language of Atkin et al. (2025), the statutory generosity of out-of-work support is not its economic generosity once the gates bind on the population actually displaced. It is worth being precise about how this sits against Dolls et al. (2012), the closest existing benchmark, because the disagreement is narrower and more interesting than a comparison against their EU averages would suggest. Their UK-specific cells (Appendix Tables 2–3) give a stabilisation coefficient of 0.352 for a proportional income shock—income tax 0.267, social insurance contributions 0.054, benefits 0.031—and 0.415 for an unemployment shock, of which income tax 0.191, SIC 0.061 and out-of-work benefits 0.163. Two things follow. First, our *levels* validate externally: 41.0 per cent on the wage-cut margin against their 35.2 for an income shock, and 32.8 per cent on displacement against their 41.5 for an unemployment shock, are the same order of magnitude computed a decade apart on different data with different software, which is a reassurance the paper is entitled to take. Second, the genuine disagreement is the *ordering*, and it localises to exactly one component. Our tax and NI components sit close to theirs on both margins; what differs is the out-of-work benefit term, 0.163 in their UK unemployment cell against 0.042 in ours (Table 3).

Most of that gap is a modelling convention rather than a change in the rules, and the paper can show as much from inside its own machinery. Dolls et al. (2012), in the standard microsimulation practice of the time, evaluate entitlement at full take-up. Our forced-100-per-cent-take-up counterfactual—the same shock, the same rules, take-up switched off as a gate in both baseline and shocked runs—raises the displacement cushioning rate from 32.9 ± 2.6 to 38.6 ± 2.5 per cent and more than doubles the UC response (Section 4.3; Appendix E), recovering most of the distance to their UK unemployment cell. That is a methodological finding in its own right, and it generalises beyond this shock: *full-entitlement microsimulation systematically overstates the automatic stabilisation of the extensive margin*, because the extensive margin is the one whose support must be claimed. The intensive margin needs no claim and no gate, so its stabilisation is measured correctly by the same convention that inflates its rival. Any stabilisation comparison across the two margins that assumes full take-up is therefore biased toward the out-of-work side by construction.

4.3 The mechanism, demonstrated

Table 3 decomposes each cushioning rate exactly. Because HBAI net income is an additive list of income, tax and benefit components, the cushioned share of the gross earnings loss splits without residual approximation into: foregone income tax, foregone employee National Insurance, additional Universal Credit, additional other benefits (JSA, ESA, housing benefit, tax credits, pension credit and the rest of the HBAI benefit list), and a remainder (zeroed employee pension contributions, statutory pay, council tax interactions).

Table 3: Component decomposition of the cushioning rate, full tariff schedule

Component (share of gross earnings loss, %)	Displacement (seed 0)	Wage cut
Income tax foregone	19.8	35.2
Employee National Insurance foregone	4.5	4.3
Universal Credit response	4.2	1.5
Other benefits (JSA, ESA, HB, tax credits, ...)	0.0	0.1
Other (pension contributions, statutory pay, ...)	4.3	0.0
Total cushioning rate	32.8	41.0

Notes: components of $1 - \Delta(\text{HBAI net income})/\Delta(\text{gross earnings})$, aggregated over all households; columns sum to the total by construction. Displacement is the seed-0 draw; the wage-cut margin is deterministic.

The decomposition overturns part of the story I would otherwise have told, and I report what it shows rather than what the prior predicted. The 8.2-percentage-point gap is more than fully accounted for by income tax: the wage-cut margin recovers 35.2 per cent of the gross loss in foregone income tax against 19.8 per cent under displacement, a 15.4-point difference that alone exceeds the whole gap. The reason is marginal-versus-average arithmetic, not means testing: a proportional wage cut shaves the *top slice* of every affected worker's earnings, relieved at marginal rates, whereas displacement removes the *entire* earnings stream—including the slice covered by the personal allowance and taxed at zero—so the tax system hands back only the average rate. National Insurance contributes equally on both margins (4.5 versus 4.3 per cent; NI's flatter schedule means its marginal and average rates differ less). Universal Credit runs the *other* way—it cushions displacement more (4.2 versus 1.5 per cent), as the naive replacement-rate intuition says it should—but far too weakly to offset the tax asymmetry, and the other-benefits row contributes nothing at all under displacement: in particular, new-style contribution-based JSA pays zero to the displaced in the model, a modelling scope limit (JSA claims are not activated by the displacement transition) that should be read alongside the UC findings below.

Why is the UC response to a mass displacement only 4 pence per pound of earnings lost? Counting displaced workers' benefit units post-shock (seed 0): 33 per cent receive positive UC, averaging £7,600 a year conditional on receipt; 67 per cent receive none (across seeds 0–4 the zero-UC share is 71 ± 16 per cent). Decomposing the zero-UC share by the constraint that binds: 55 per cent of all displaced workers are in families modelled as not taking up UC (PolicyEngine's stochastic take-up draw, held fixed from the baseline), 12 per cent are in families whose joint means test tapers the maximum award to zero against a partner's earnings and unearned income (18 ± 9 per cent across seeds), and—contrary to the mechanism the working-paper draft asserted—0 per cent are excluded by the £16,000 capital limit. The capital result is a data limitation, not a finding about the rules: the plain FRS 2024–25 dataset used here carries no household capital (the UC-countable capital variables are zero throughout), so the capital test *cannot* bind in the model, and this paper cannot test how many displaced workers it would disqualify in reality. A counterfactual rerun makes the same point mechanically: neutralising the UC capital rules (raising the £16,000 limit and the £6,000 tariff-income threshold by five orders of magnitude, taper and work allowances unchanged) leaves the displacement cushioning rate identical to four decimal places (32.83 per cent), where the asserted mechanism predicted it would rise toward the wage-cut margin's 41 per cent. The take-up gate, by contrast, is quantitatively live: forcing 100 per cent UC take-up in both the baseline and shocked simulations (the plain FRS dataset's modelled take-up flags imply 55 per cent of individuals are in benunits that would claim) more than doubles the UC response to displacement (£0.16 versus £0.07 billion a year) and raises the displacement cushioning rate from 32.9 ± 2.6 to 38.6 ± 2.5 per cent over 10 common draws—closing about seventy per cent of the gap to the wage-cut margin's 41 per cent (Appendix E).

The mechanism itself should be claimed carefully, because it is older than this paper. That the income tax stabilises at effective *marginal* rates is the organising idea of Auerbach and Feenberg (2000); McKay and Reis (2016) formalise how progressivity and the intensive/extensive structure of a shock determine stabilisation in general equilibrium; and Dolls et al. (2012) explicitly read their own coefficients as effective marginal rates and find the tax term larger for the income shock than for the unemployment shock in all nineteen EU countries and the United States. The marginal-versus-average asymmetry is thus a known, universal feature, and we do not claim it. What is new here is threefold. First, the *sign flip of the total*: everywhere in Dolls et al. (2012) the always-present tax asymmetry is outweighed by out-of-work support, so the unemployment shock is the better-stabilised one; in the UK of the 2020s, with out-of-work support having shrunk relative to earnings and gated by take-up, the tax asymmetry now dominates and the ordering reverses. Second, the decomposition here is exact and residual-free, computed on the population *actually displaced* by an identified shock rather than on a proportional shock applied to everyone—so the average rates in the denominator are the average rates of real manufacturing workers, not of the workforce. Third, the gate-by-gate audit: we do not infer that UC fails to reach the displaced, we count which constraint binds on each family and report that take-up accounts for 55 per cent of the failures, the partner-income taper for 12, and the capital test—the mechanism an earlier draft asserted—for none on these data.

The corrected reading of the headline result is therefore: the tax–benefit system cushions wage cuts more than job losses *first* because income tax relieves marginal earnings at marginal rates but lost jobs only at average rates, and *second* because the extensive margin of UC is gated—by modelled take-up and by the joint partner-income taper on these data; plausibly also by the capital test in reality, but that channel is unobservable in the plain FRS and contributes nothing here. The qualitative claim that survives intact is that out-of-work support reaches only a third of the displaced while in-work tax relief reaches every wage-cut worker automatically; the specific attribution to the capital means test does not.

4.4 Distributional incidence by adjustment margin

Figure 1 shows the mean change in household disposable income by equivalised income decile under the three original margins, full schedule, Monte Carlo means over the 50 draws with ±1 SD bands. Two profiles emerge. The displacement margin is top-half concentrated: mean losses are largest in deciles 7–10 (£71–174 per person per year, with draw-to-draw SDs of £43–135 in those deciles—individual draws are lumpy, reflecting which households happen to contain the drawn workers) and small in the bottom four deciles (under £20). The wage-cut margin is smooth and rises broadly with income, from £6 in the bottom decile to £116 at the top (the profile is not monotone—it dips slightly at deciles 2, 8 and 9, reflecting the industry composition of those deciles' earners—but the gradient across the distribution is unambiguous): exposed manufacturing earnings sit disproportionately in the upper-middle of the distribution, and proportional cuts scale with earnings.

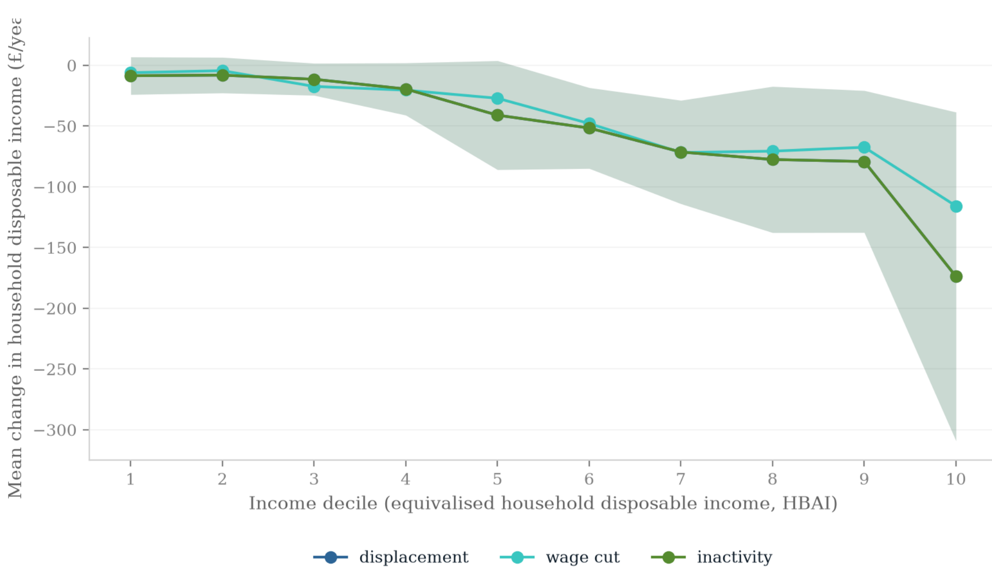

Figure 1: Mean change in household disposable income by equivalised income decile under the three original adjustment margins, full tariff schedule. Lines are Monte Carlo means over 50 displacement draws; shaded bands are ±1 draw-to-draw SD (the wage-cut margin is deterministic). The inactivity margin tracks displacement closely by decile; its LCWRA health element moves the aggregate cushioning and fiscal numbers rather than the decile profile (see text).

This decile profile yields a named finding: *the tariff shock is regionally, not vertically, concentrated*. Displacement losses land hardest on deciles 7–10—manufacturing employees are mostly not poor—so the aggregate inequality and poverty statistics barely move. The Gini coefficient of equivalised disposable income rises by 0.00022 ± 0.00017 under full displacement (from a baseline of 0.309), and actually *falls* very slightly (-0.0001) under the wage-cut margin, since proportional cuts to above-median earnings are mildly equalising. Relative BHC poverty rises by 0.055 ± 0.021 percentage points under full displacement—roughly 38,000 people—and by essentially zero under the wage-cut margin: a several-per-cent wage cut pushes almost no exposed household across the poverty line, while a job loss pushes some. Absolute BHC poverty against a fixed baseline line (computed manually at seed 0; Appendix C) moves nearly identically (+0.058 points full displacement, +0.029 EPD, zero under wage cuts), confirming that the relative-line result is not an artefact of a shock-shifted median.

The near-null national headcount conceals an intensive margin that the poverty *gap* makes visible. The aggregate relative BHC poverty gap—the weighted sum of household shortfalls below the line—rises by £0.19 billion a year ± 0.05 under full-schedule displacement (£0.14 billion ± 0.06 under the EPD) on a national baseline gap of £36.3 billion: a 0.5 per cent movement, invisible nationally, and only £0.014 billion (deterministic) under the wage-cut margin. But essentially the entire gap increase is borne inside *affected* households (those containing a displaced worker; 20 draws, seeds 0–19): their own baseline poverty gap of £0.01 billion rises twenty-fold to £0.19 billion post-shock, and the relative BHC poverty rate among the roughly 94,000 people in those households jumps from 4 ± 6 to 43 ± 13 per cent—a rise of 39 ± 12 percentage points (34 ± 13 points under the EPD). Among the 4.6 million people in wage-cut-affected households, by contrast, the poverty rate does not move at all. The correct reading of the "+0.055 points" headline is therefore not that the shock is poverty-irrelevant, but that a full-year displacement is close to poverty-*certain* for the small population it hits, while remaining invisible in national statistics. The distributional action, such as it is, therefore runs along the margin (who loses work versus who loses wages) and along geography (Section 4.8)—not down the income distribution.

The inactivity margin is substantively distinct from displacement: displaced over-50s transition to economic inactivity *and* their Universal Credit includes the LCWRA health element (Section 3.3), under the stated upper-bound assumption that all of them pass the Work Capability Assessment. The three-way comparison under the full schedule (inactivity: 20 draws, seeds 0–19) is: cushioning rates of 32.8 per cent (displacement, seed 0), 35.3 ± 4.2 per cent (inactivity) and 41.0 per cent (wage cut); Exchequer costs of £0.85 billion ± 0.30 (displacement), £0.81 billion ± 0.26 (inactivity) and £0.98 billion (wage cut); relative BHC poverty changes of $+0.055 \pm 0.021$, $+0.055 \pm 0.017$ and 0.00 percentage points. The inactivity–displacement contrast is a paired-seeds statement: on common seeds the health element makes inactivity *more* cushioned and *more* expensive than displacement—by about £15 million a year, positive in 19 of 20 paired draws under the full schedule and 20 of 20 under the EPD—while the unpaired 20-draw inactivity mean sits below the 50-draw displacement mean purely through draw-set noise. The wedge is small because the element pays only where UC is actually in payment: roughly 11,800 of the 33,600 displaced are 50 or over and flagged, but two-thirds of displaced families clear no UC gate (Section 4.3), so most LCWRA elements attach to awards that are already zero. Poverty is essentially unchanged (marginally *lower* under inactivity in 7 of 20 draws, never higher), and no cross-margin ordering flips: among the three earnings-equivalent families displacement remains the least cushioned and cheapest, wage cuts the most cushioned and most expensive, and benefit-financed inactivity sits strictly between (the reallocation family of Section 4.9 removes a smaller gross loss, so it is comparable on rates but not on levels). Under the 50-per-cent-WCA-pass sensitivity variant the seed-0 wedge roughly halves (Exchequer £0.72 billion against £0.73 billion at full take-up and £0.71 billion under pure displacement; cushioning 33.2, 33.6 and 32.8 per cent). The remaining caveats are assessment realism—no WCA process, waiting period or conditionality regime is modelled—and the upper-bound qualification (Section 6); UK history says the margin matters (Beatty et al., 2007; Beatty and Fothergill, 2017), and the fiscal wedge quantified here is its first-round price.

4.5 Who bears it: gender, age and household type

Tables 4 and 5 report who the displaced are and how much of their loss the state absorbs, over 20 Monte Carlo draws (seeds 0–19; full schedule). Group cushioning rates attribute each household's disposable-income change to its affected workers in proportion to their gross earnings losses, so the rows aggregate exactly to the economy-wide rates of Table 2.

Three patterns are worth naming, with the standing Monte Carlo discipline that group-level draws are lumpy. First, gender: 73±9 per cent of the displaced are men—exposed manufacturing is male-dominated—but the system cushions displaced women *less* (28 ± 7 against 35 ± 5 per cent), a one-sided contrast in almost every draw even though the means sit within one joint SD. Under the wage-cut margin the gender gap in cushioning nearly closes (40.4 versus 41.2 per cent): the asymmetry is a property of the gated out-of-work system, not of the tax schedule. This is the tax–benefit-system counterpart of the De Lyon and Pessoa (2021) finding that UK women bore the China shock disproportionately through lost employment years: here, when women do lose manufacturing jobs, the state also replaces less of the loss. Second, age: displacement is concentrated in prime age (30 and 25 per cent of the displaced are 35–44 and 45–54 respectively), and cushioning is weakest at the two ends—29 ± 4 per cent for under-25s and 27 ± 12 per cent for the 65-plus—the groups least likely to clear UC's gates. Third, household type, which is where the mechanism section's partner-income taper becomes visible in the incidence: half the displaced (50 ± 14 per cent) live in dual-earner couples, and their cushioning rate (32.5 ± 4.9 per cent) sits below single-earner couples' (36.4 ± 7.6)—directionally the taper gate at work (a working partner's earnings exhaust the UC award), though the two means sit within one SD of each other and I flag rather than lean on the pairwise ordering. The cleaner statement is again the cross-margin one: under wage cuts, where UC is largely irrelevant, dual- and single-earner couples are cushioned almost identically (42.0 versus 43.1 per cent); the differential opens only on the displacement margin, where the joint means test operates.

4.6 Fiscal incidence

Table 6 reports the annual Exchequer cost—foregone tax plus additional benefit spending—by scenario, and Figure 2 shows the full draw distribution. Under the full schedule, displacement costs £0.85 billion $\pm$ 0.30 (mean $\pm$ SD over 50 draws) and the wage-cut margin £0.98 billion; under the EPD, £0.66 billion $\pm$ 0.33 and £0.77 billion. The wage-cut margin costing *more* than displacement is the fiscal mirror of the cushioning result: the Exchequer absorbs wage cuts automatically at marginal income-tax rates for every affected worker, whereas a displacement destroys tax liability only at average rates and—with two-thirds of displaced families receiving no UC (Section 4.3)—triggers comparatively little benefit spending. That said, MC discipline applies: the displacement–wage-cut gap of £0.13 billion (full schedule; £0.11 billion EPD) is smaller than the £0.30 billion draw-to-draw standard deviation, and the deterministic wage-

Table 4: Displacement incidence by gender and age band, full tariff schedule

	Displacement (20 draws)			Wage cut
Group	Share of displaced (%)	Mean income change per displaced (£/yr)	Cushioning rate (%)	Cushioning rate (%)
Men	73 ± 9	$-37{,}900 \pm 8{,}800$	35 ± 5	41.2
Women	27 ± 9	$-28{,}600 \pm 9{,}800$	28 ± 7	40.4
16–24	7 ± 4	$-27{,}200 \pm 9{,}800$	29 ± 4	29.6
25–34	19 ± 8	$-26{,}800 \pm 7{,}500$	30 ± 8	37.8
35–44	30 ± 12	$-43{,}100 \pm 22{,}400$	33 ± 7	42.9
45–54	25 ± 7	$-34{,}000 \pm 8{,}700$	34 ± 6	45.2
55–64	17 ± 10	$-31{,}100 \pm 12{,}000$	33 ± 10	39.1
65+	7 ± 6	$-41{,}200 \pm 66{,}000$	27 ± 12	34.8

Notes: Monte Carlo means ± SD over 20 displacement draws. Income changes are the household HBAI net-income change attributed to affected workers in proportion to their gross earnings losses, per displaced worker per year; group cushioning is 1− attributed net loss / gross earnings loss. The wage-cut column is the deterministic run's group cushioning rate (its per-worker income changes, £400–900 a year, are on a different scale and omitted).

Table 5: Displacement incidence by household type, full tariff schedule

	Displacement (20 draws)			Wage cut
Household type	Share of displaced (%)	Mean income change per displaced (£/yr)	Cushioning rate (%)	Cushioning rate (%)
Single, no children	24 ± 9	$-28{,}700 \pm 6{,}100$	33 ± 5	36.9
Single, with children	6 ± 3	$-30{,}900 \pm 20{,}600$	33 ± 14	44.0
Couple, no children, single earner	12 ± 8	$-36{,}300 \pm 23{,}800$	34 ± 10	40.2
Couple, no children, dual earner	24 ± 11	$-35{,}200 \pm 9{,}500$	29 ± 6	38.5
Couple, with children, single earner	10 ± 5	$-49{,}000 \pm 43{,}900$	34 ± 10	45.8
Couple, with children, dual earner	26 ± 13	$-35{,}500 \pm 9{,}200$	34 ± 6	45.2
All couples, single earner	21 ± 11	$-43{,}500 \pm 27{,}000$	36 ± 8	43.1
All couples, dual earner	50 ± 14	$-34{,}900 \pm 6{,}200$	33 ± 5	42.0

Notes: as Table 4. Household type is the displaced worker's benunit family type at baseline; earner counts are baseline counts of benunit adults with positive employment income. The bottom panel pools couples across children status.

cut cost exceeds the displacement draw in 37 of 50 draws under each schedule—suggestive and mechanism-consistent, but not sharply identified by the draws alone; the cushioning rates of Table 2, which compare aggregates rather than draws, carry the identification. Orderings that clearly survive the SD: every scenario costs the Exchequer strictly more than zero. The full schedule costs more than the EPD within each margin in expectation, but this too is a paired-draws statement: the full−EPD difference on common seeds averages £0.19 billion ± 0.38 and is positive in 36 of 50 paired draws—directionally consistent, though individual draws can reverse it.

Table 6: Annual Exchequer cost by scenario (2026, £bn)

	Displacement	Wage cut	Inactivity
Full tariffs	0.85 ± 0.30	0.98	0.81 ± 0.26
EPD	0.66 ± 0.33	0.77	0.62 ± 0.26

Notes: displacement means ± SD over 50 Monte Carlo draws; inactivity over 20 draws (seeds 0–19) with the LCWRA health element; the wage-cut margin is deterministic. Unpaired inactivity means sit below the 50-draw displacement means through draw-set noise; on common seeds inactivity costs about £0.015bn *more* than displacement, positive in 19 of 20 (full) and 20 of 20 (EPD) paired draws (Section 4.4). Fiscal policy held fixed; no revenue recycling (Ignatenko et al., 2025).

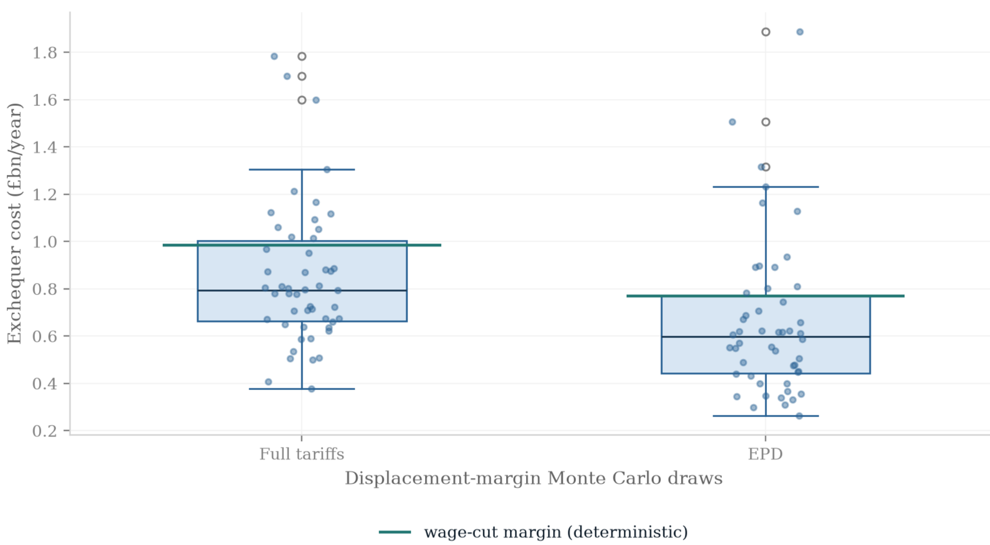

Figure 2: Draw distribution of the annual Exchequer cost on the displacement margin: box plots and individual draws over the 50 Monte Carlo displacement draws, full tariffs and EPD, with the deterministic wage-cut cost overlaid as a horizontal line. The wage-cut cost sits above the displacement draw in 37 of 50 draws under each schedule, but inside the interquartile whisker range—the ordering is suggestive, not sharply identified by the draws.

The displacement costs above annualise a full-year out-of-work spell; they scale roughly pro rata with the expected duration. Rerunning displacement with displaced workers keeping half a year's earnings (a six-month expected spell, evaluated in-model so taxes and benefits respond to the actual annual income) cuts the Exchequer cost from £0.70 to £0.41 billion over 10 common draws—slightly *more* than half, because the retained half-year of earnings is taxed at rates above the displaced worker's average rate—and lifts the cushioning rate from 33 to 42 per cent, while the poverty effect nearly vanishes (+0.013 against +0.056 percentage points): half a year of earnings keeps most displaced households above the line (Appendix E).

Whichever margin, the fiscal incidence dwarfs the poverty incidence. Set against a poverty rise of at most six-hundredths of a percentage point, an annual Exchequer cost of £0.7–1.0 billion says the 2025 tariff reaches the UK household sector primarily as a public-finance problem, mediated by the automatic stabilisers, rather than as a poverty problem.

4.7 The EPD counterfactual

Table 7 and Figure 3 price the Economic Prosperity Deal as the difference between the two tariff scenarios run through identical machinery. On the displacement margin the deal averts about 6,500 job losses: that is the mean paired-draw difference over the 50 Monte Carlo displacement draws (34,211 versus 27,700, ±4,388, positive in 47 of 50 paired draws), and it is the figure this paper quotes throughout. The corresponding quota expectation—the analytic head-count target before any draw is realised—is 6,857 (33,566 versus 26,709). The deal also averts £0.38 billion of gross annual earnings loss (£1.77 versus £1.39 billion), and roughly £0.2 billion of annual Exchequer cost (£0.19 billion ± 0.38 mean paired-draw difference on the displacement margin, positive in 36 of 50 paired draws; £0.22 billion, deterministic, on the wage-cut margin). Three sectoral clauses do all the work. The auto rate cut from 25 to the in-quota 10 per cent more than halves the largest single-sector shock, and because the 100,000-vehicle quota approximately equals 2024 US export volumes, the in-quota rate is the operative one. The pharmaceutical exemption removes the SIC 21 employment shock entirely—by construction, since the exemption's price was paid elsewhere (Section 5.2). Steel relief is modelled as partial (half the 25 per cent rate), and carries the standing caveat that with roughly 80 per cent of UK steel exports going to the EU, the US-tariff cell understates the sector's true trade-policy exposure, which is dominated by EU/UK safeguards.

Table 7: What the EPD averted: full tariffs versus EPD, annual

	Full tariffs	EPD	Averted
Displaced workers			
Quota expectation	33,566	26,709	6,857
Monte Carlo mean	$34{,}211 \pm 3{,}600$	$27{,}700 \pm 4{,}800$	$6{,}511 \pm 4{,}388$
Annual totals (£bn)			
Gross earnings loss	1.77	1.39	0.38
Exchequer, displacement	0.85 ± 0.30	0.66 ± 0.33	0.19 ± 0.38
Exchequer, wage cut	0.98	0.77	0.22
Relative BHC poverty (ppt)	$+0.055 \pm 0.021$	$+0.042 \pm 0.022$	0.013

Notes: the first row is the analytic quota expectation—the head-count target implied by s_j before any draw is realised. The second is the Monte Carlo mean over 50 displacement draws; its "averted" column is the mean *paired-draw* difference on common seeds, positive in 47 of 50 paired draws. This 6,511 figure (quoted as "about 6,500") is the paper's headline count. Averted Exchequer costs are likewise mean paired-draw differences on common seeds (± SD of the paired difference; positive in 36 of 50 draws). The wage-cut margin is deterministic.

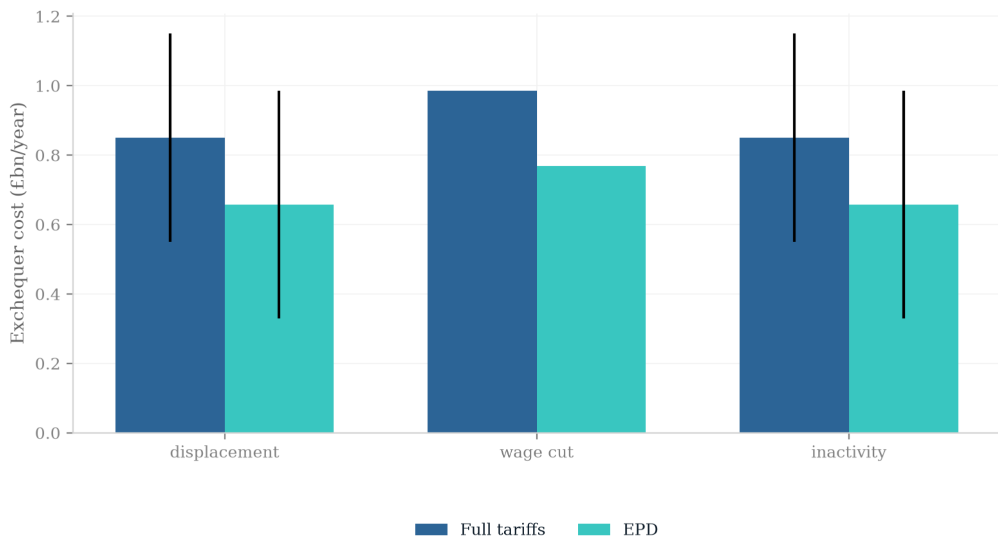

Figure 3: Annual Exchequer cost, full tariffs versus EPD, by adjustment margin. Bars are means over 50 draws; error bars are draw-to-draw standard deviations (wage cut is deterministic).

4.8 Regional incidence

The regional profile is where the shock's concentration actually lives, and it moves with the margin. The main-text geography is the ITL1/government office region level, because that is the level at which the FRS records region as an *observed* variable; the map is Figure 4 and the full three-scenario table is Table 13, both in Appendix G. A constituency-level projection is reported alongside them, as an explicitly illustrative exhibit (Appendix G.2)—it rests on imputed industry codes and cannot reproduce observed local industry mix, so it is not carried in the main text. All regional displacement figures below are Monte Carlo means ± SD over the 50 draws. Under full-schedule displacement the largest mean regional losses are Wales (£108 ± 234 per person per year—the enormous SD reflects a single high-weight Welsh record that dominates the draws it enters), the East of England (£74 ± 76), the East Midlands (£72 ± 73, reflecting machinery, transport-equipment and food-manufacturing employment), the West Midlands (£64 ± 56) and Scotland (£62 ± 50, with beverages, i.e. Scotch whisky at a 6.3 per cent US-export intensity, prominent). No pairwise regional ordering survives one draw-to-draw SD, and I do not assert one. Under the deterministic wage-cut margin the same aggregate loss spreads smoothly: Wales (£65), the East of England (£61) and the West Midlands (£55) lead, because the proportional cut weights regions by their exposed *wage bills* rather than by where the discrete displacement draw lands. Under EPD displacement the mean profile is led by Wales (£75 ± 181), the East Midlands (£70 ± 93) and Northern Ireland (£63 ± 52); Yorkshire and the Humber, the steel region for which the deal's conditional half-relief is the only mitigation, averages £34 ± 37—though its worst single draw reaches £184 under the EPD (and £209 under the full schedule), which illustrates exactly how far one draw can sit from the mean. The concentration signature is therefore a statement about *within-draw* geography: in any single draw the displacement losses pile up in a handful of regions—which is why the region-level SDs are comparable to or larger than the means—while wage cuts spread thinly everywhere. That is the margin's geographic signature, and it is the regional counterpart of the Gini result: a shock that is invisible in national inequality statistics can still be a £100–200-per-head event in whichever regions a given realisation hits.

4.9 The reallocation margin

The three families of Section 3.3 leave a gap that the China-shock literature fills empirically: displaced manufacturing workers do not, for the most part, stay out of work. They move into services—health and social care, retail, delivery and land transport, food service—at a wage penalty (Autor et al., 2014; Dauth et al., 2021; Traiberman, 2019). Autor et al. (2014) track US workers' cumulative earnings after trade exposure and find the losses run through re-employment at lower pay rather than through permanent non-employment; Dauth et al. (2021) and Traiberman (2019) document the same reallocation-with-penalty pattern in German and Danish administrative data. A margin grid that prices only job loss, an in-place wage cut and inactivity therefore omits the case the evidence actually describes. This subsection adds it: a fourth family in which the *same* drawn workers as the displacement family keep a job, but in services and at a calibrated earnings penalty.

Calibration. The penalty is measured on the FRS 2024–25 itself at period 2026, grossing-weighted over employees with positive earnings. The source population is the exposed goods-producing divisions SIC 10–32—mean annual earnings £48,272 over 2,047 annual hours, £23.58 an hour. The destination population is the four services divisions the literature identifies as absorbing displaced manufacturing labour—SIC 47 retail, 86 human health, 49 land transport (including delivery and couriers) and 56 food and beverage service—mean annual earnings £34,594 over 1,701 hours, £20.34 an hour. The ratio implies a **28.3 per cent annual-earnings penalty**, and it deliberately embeds the hours fall (1,701 against 2,047), because annual earnings, not the hourly wage, is what enters household income and the tax–benefit calculation. Controlling crudely for hours and age (weighted OLS of log annual earnings on a destination dummy, log hours, age and age squared over the same population) leaves a **14.0 per cent** pure hourly-wage penalty—the unconditional hourly gap is 13.8 per cent—and that figure is run as an explicit low-penalty variant rather than buried as a footnote. ASHE 2024 cross-checks the range: full-time median gross weekly pay is about £700 in manufacturing against £590 in retail, £650 in transport and storage, £670 in health and £490 in food service, a 10–20 per cent full-time-only gap that brackets the hours-controlled figure. The FRS all-employee number is larger precisely because it also captures the shift into part-time work, which the full-time-only ASHE comparison excludes by construction. Destination assignment is stochastic in proportion to FRS employee-weight shares within the four divisions: health 45.8 per cent, retail 29.5, food service 15.5, land transport 9.2. A variant places workers out of work for three months before the services job starts, scaling annual earnings and hours by a further $1 - 3/12$.

Results. Table 8 sets the four families side by side under the full schedule, with the lag and low-penalty variants. The reallocation runs use 20 draws with seeds $0, \ldots, 19$, *shared with the displacement and inactivity runs*, so every reallocation–displacement comparison is paired draw-for-draw: within a given seed the same workers are hit in every family, 33,566 weighted on average across the 20 draws.

On the **cushioning rate**—the scale-free object—reallocation lands exactly where the mechanism of Section 4.3 predicts, between the two poles: 0.3445 (displacement) $<$ 0.3919 (reallo-

Table 8: The four adjustment margins, full tariff schedule, period 2026

Margin	Gross loss (£bn)	Exchequer cost (£bn)	Rel. BHC poverty (ppt)	Cushioning rate
Displacement	1.823	0.793 ± 0.261	$+0.058 \pm 0.019$	0.3445 ± 0.038
Inactivity	1.823	0.808 ± 0.264	$+0.055 \pm 0.017$	0.3527 ± 0.042
Reallocation (28.3%)	0.516	$\mathbf{0.277 \pm 0.077}$	$\mathbf{+0.005 \pm 0.007}$	$\mathbf{0.3919 \pm 0.029}$
+ 3-month lag	0.843	0.442 ± 0.128	$+0.009 \pm 0.008$	0.3795 ± 0.030
low penalty (14.0%)	0.255	0.139 ± 0.038	$+0.002 \pm 0.005$	0.4010 ± 0.029
Wage cut	1.771	0.985	+0.000	0.4102

Notes: Monte Carlo means ± draw-to-draw SD over 20 draws, seeds $0, \ldots, 19$, shared across the displacement, inactivity and reallocation families so that differences are paired. The wage-cut margin is deterministic. Within a seed all families hit the same workers (33,566 weighted on average across draws). *The gross-loss column is not constant across families and must be read before the Exchequer column*: the reallocation family removes only the wage penalty on movers ($0.283 \times 1.823 = 0.516$), whereas the displacement and wage-cut families remove the full earnings-equivalent aggregate. Cushioning rate is $1 -$ (net disposable loss/gross earnings loss), so it is scale-free and comparable across all rows; the pound columns are comparable only within the displacement–reallocation block.

cation) < 0.4102 (wage cut). The ordering is monotone in time spent out of work: inserting a three-month jobless spell pulls the reallocation rate back towards displacement (0.3795), and halving the penalty pushes it towards the wage-cut pole (0.4010). The reason is the paper's central mechanism: a worker who keeps an earnings stream has their loss relieved at marginal income-tax rates and through in-work Universal Credit tapers, while a worker who loses the job entirely is relieved only at average rates and must pass through the gated extensive margin of the benefit system. Under the EPD schedule the ordering is identical: 0.3464 (displacement) < 0.3684 (lag-3) < 0.3772 (reallocation) < 0.3882 (low penalty) < 0.4079 (wage cut), with Exchequer costs of £0.606, 0.327, 0.204 and 0.103 billion against the wage cut's £0.769 billion, and poverty changes of +0.039, +0.010, +0.005 and +0.002 percentage points against +0.000.

An ordering that must not be over-read. On *levels* reallocation is not intermediate—it is the cheapest family in the paper, and the reason is arithmetic, not economics. The reallocation family removes only the wage penalty on the workers who move ($0.283 \times$ £1.823 billion = £0.516 billion of gross earnings), whereas the wage-cut family is calibrated to remove the full earnings-equivalent aggregate spread across the entire division wage bill (£1.771 billion). The two families are answering different questions and their pound figures are not commensurable. Reallocation is level-comparable *only* with displacement, where the worker sets and the draws are identical; against the wage cut it is a comparison of rates, not of costs. I state the honest summary directly: *conditional on the same workers being hit, landing a services job at the FRS-measured penalty removes about two-thirds of the fiscal cost and roughly nine-tenths of the poverty impact of job loss.* The paired-draw statistics make that precise. Relative to displacement on common seeds, reallocation lowers the Exchequer cost by £0.516 billion (SE of the paired difference 0.042), lowers the relative BHC poverty rise by 0.0525 percentage points (SE 0.0046) and raises the cushioning rate by 0.0474 (SE 0.0056); with a three-month lag the differences are −£0.351 billion (SE 0.031), −0.0488 percentage points and +0.0350 (SE 0.0044). Under the EPD they are −£0.402 billion (SE 0.041), −0.0335 percentage points (SE 0.0033) and +0.0308 (SE 0.0081) for reallocation, and −£0.279 billion (SE 0.031), −0.0290 percentage points and +0.0220 (SE 0.0064) with the lag. Every one of these is many standard errors from zero—the pairing on shared seeds removes almost all of the draw noise that makes the level SDs so large.

What the model does and does not price. A second caveat deserves the same plainness. The sector switch is literal: each reallocated worker's `sic_industry_division` is reassigned to one of the four destinations, and the assignment is reported. But `sic_industry_division` is a pure input variable in PolicyEngine UK—nothing downstream in the tax–benefit calculation reads it. The destination therefore carries no fiscal content of its own, and *all* of the economic content of this family comes from the earnings and hours penalty applied to the mover. Nothing here should be read as the model pricing the sector of re-employment; it prices a calibrated earnings cut on a set of workers who would otherwise have been unemployed.

Inequality. The Gini result reinforces the paper's existing scoping of distributional claims. Reallocation leaves the Gini of equivalised disposable income essentially unmoved (−0.00001), against +0.00027 under displacement. The inequality signature of this shock, small as it already is, is a property of the *displacement* margin specifically—of workers leaving the earnings distribution altogether—and not of the tariff shock as such. Section 6 scopes the two structural distributional claims accordingly.

4.10 Validation against the 2025–26 outturn

The design lets the realised data discipline the one free parameter. ONS monthly trade data show UK goods exports to the US falling 24.7 per cent in April 2025, the first full tariff month; against the 12.8 per cent trade-weighted average tariff implied by the primitives in Table 1, this pins the export-demand elasticity at 1.93, which I round to 2.0. The calibration is deliberately an *outturn* anchor rather than a literature import—but it shares April's observation, so it validates nothing by itself. The out-of-sample test uses the ten subsequent months. From the same HMRC customs source as the intensity build, pulled monthly (Appendix A), I compute each division's realised export fall over May 2025 to February 2026 against the mean of the same calendar months one and two years earlier—same-month comparisons net out seasonality, and the two-year average dilutes the front-running contamination of the one-year base; April 2025, the calibration month, is excluded.

Because the paper quotes both a 24.7 and a 12.7 per cent fall, the relation between them should be stated plainly rather than left to the reader: they are different statistics, not competing estimates, and nothing here contradicts the ONS. The 24.7 is an April 2025 month-on-month fall against the front-run March peak; the 12.7 is a ten-month window against a two-year same-calendar-month baseline. Computed on the ONS's own month-on-month basis, our data give 21.7 per cent for April (£4.85 billion to £3.80 billion), with the remaining three points accounted for by the HMRC customs versus ONS balance-of-payments basis—our 2024 mapped total is £52.5 billion of a £55.6 billion all-commodity customs total against an ONS balance-of-payments figure of about £59.3 billion—rather than by any difference in the underlying movement. Both series exclude non-monetary gold and the other precious metals, and that exclusion is load-bearing: on the same April calculation, retaining them would report a 44.2 per cent fall, which is exactly the bullion-driven artefact the exclusion exists to remove (Appendix A). Nor is the window result an artefact of baseline construction—every post-window month falls in the £3.34–4.22 billion range against a 2024 monthly range of £4.07–4.83 billion, so every post month sits below every 2024 equivalent.

Table 9 sets the ε-model's predicted export falls ($\varepsilon\tau_j$, full and EPD schedules) against the realised falls for the ten divisions with the largest derived full-schedule shocks. Three honest readings. First, the *aggregate* holds up reasonably: mapped US exports fell 12.7 per cent over the window, about half the April collapse persisting once the EPD (in force for autos from late June 2025) and the post-April rebound are in the data. Against trade-weighted aggregate predictions of 25.1 per cent (full schedule) and 17.4 per cent (EPD schedule), the realised 12.7 sits below both but far nearer the EPD—the right ordering for an outturn that embeds the deal, with the residual gap the footprint of rebound and re-routing that a static demand curve does not capture. The autos-led profile is confirmed: motor vehicles show the largest realised fall among the schedule-differentiated divisions (32.0 per cent, against a predicted 50 under the full schedule and 20 under the EPD)—though that figure is baseline-dependent in the same way as pharmaceuticals below: on a calendar-2025-versus-2024 baseline the car fall is 27.4 per cent, and the ONS reports 24.4, so the two-year baseline, which deliberately dilutes the front-run base, is itself the main source of the gap to the published figure. That is a defensible choice for a measure meant to capture the tariff rather than the anticipation of it, but it does deepen the measured fall and should be read as such. Second, the *cross-section* is where the model is weakest, and the table says so: the Spearman rank correlation between predicted and realised falls across the top ten is 0.00 under the full schedule and, on the adversarial read that deserves the main text rather than a table note, −0.62 under the EPD: across the ten most exposed divisions the deal-mitigated schedule ranks them not merely uninformatively but backwards. The failure is informative about channels the tariff schedule does not encode. Pharmaceuticals fell *despite* the eventual exemption—the Section 232 threat and the unwinding of the enormous Q1-2025 stockpiling operated on a division the EPD schedule scores at zero—though the size of that fall is the most baseline-sensitive number in the paper and is better reported as a span than a point. The crosswalk is exact here (division 21 is SITC 54, identical to the ONS's "medicinal and pharmaceutical products"), so the divergence from the ONS's published 16.2 per cent is not a mapping problem but a baseline choice: our own data give 10.9 per cent on a calendar-2025-versus-2024 basis and 30.7 per cent on the two-year-average baseline, with the ONS figure sitting inside that range. The honest statement is a realised pharma fall of 10.9 to 30.7 per cent, with 30.7—the figure that enters the measured family—the upper end of that span rather than the estimate. Wearing apparel fell 28.9 per cent on a flat 10 per cent tariff. Exports *rose* in aerospace (+9.8 per cent; the EPD's aerospace carve-out, which the stylised schedule leaves at 10 per cent), in basic metals (+11.2 per cent) and in electrical equipment. Third, at the level that actually enters the microsimulation—the earnings shock s_j, which multiplies the fall by the export *intensity*—the full-schedule model and the measured build agree better: Spearman 0.58 across all 23 divisions, because the intensity term, which is measured rather than predicted, carries most of the cross-sectional variance. The fair summary is that the one-parameter model gets the aggregate and the autos signature right and the within-manufacturing ranking wrong; the measured family of Section 4.11 therefore replaces the prediction with the outturn wholesale.

The basic-metals rise deserves separating out, because it is the one apparent anomaly in the table that turns, on inspection, into a substantive point in the model's favour. The +11.2 per cent is not a precious-metals artefact—non-monetary gold and the silver and platinum group are excluded throughout (Appendix A)—but a composition effect within a division that mixes tariffed and untariffed products. Inside division 24 every ferrous steel group moves as the tariff predicts: SITC 671 falls 21 per cent, 672 by 20, 673 by 60, 674 by 5, 677 by 19 and 679 by 27. The division-level rise comes entirely from non-ferrous metals lying outside the steel tariff lines—copper (682) up 72 per cent, lead (685) up 134, and alloy and wire products (675, 676) up 105 and 37 respectively—with copper separately front-run ahead of its own Section 232 action. The division therefore averages a tariffed steel collapse with an untariffed and partly anticipatory non-ferrous surge, and the division-level figure *understates* the steel shock rather than contradicting it. We nevertheless do not adjust it, for a unit-of-analysis reason: the division is the level at which employment counts and the FRS crosswalk are defined, so a sub-division reweighting of the export fall would have no matching denominator on the employment side and would silently mix units. A similar check clears the other manufacturing division: SIC 32's realised fall is not precious-metal-driven either, since SITC 897 (articles of precious metal) is only £187 million of post-window flow and itself falls 24.6 per cent.

Table 9: Predicted versus realised US-export falls, top-10 exposed divisions

SIC	Division	Predicted fall $\varepsilon\tau_j$ (%) Full	EPD	Realised fall (%) May 2025–Feb 2026
29	Motor vehicles	50.0	20.0	32.0
28	Machinery and equipment	20.0	20.0	11.3
26	Computer, electronic, optical	20.0	20.0	4.8
30	Other transport (incl. aerospace)	20.0	20.0	−9.8
14	Wearing apparel	20.0	20.0	28.9
32	Other manufacturing	20.0	20.0	15.6
21	Pharmaceuticals	20.0	0.0	30.7
27	Electrical equipment	20.0	20.0	−1.7
24	Basic metals (incl. steel)	50.0	25.0	−11.2
20	Chemicals	20.0	20.0	14.5
Aggregate (all mapped divisions)				12.7
Spearman rank correlation, top 10		0.00	−0.62	

Notes: divisions ordered by the derived full-schedule shock s_j. Predicted falls are $\varepsilon\tau_j$ with $\varepsilon = 2.0$. Realised falls are HMRC customs-basis UK goods exports to the US, May 2025–February 2026, against the mean of the same calendar months one and two years earlier (Appendix A); negative entries are export *rises*. Two rows are materially baseline-sensitive and should be read as spans: pharmaceuticals is 10.9 to 30.7 per cent (ONS 16.2) and motor vehicles 27.4 to 32.0 per cent (ONS 24.4) across the calendar-year and two-year-average baselines; the two-year baseline reported here dilutes the front-run base and so gives the deeper fall. Basic metals is a composition effect, not a precious-metals artefact: the ferrous steel groups within division 24 all fall, and untariffed non-ferrous metals drive the division-level rise. Rank correlations are Spearman coefficients between each predicted column and the realised column over the ten rows. At the earnings-shock level $s_j = (\text{fall}_j)x_j$, the full-schedule model and the measured build correlate at 0.58 (Spearman, all 23 divisions). Full division-level detail is reported in the replication materials.

Which *adjustment margin* the data chose remains harder to read, and this paper does not resolve it. The natural instrument is the ONS Business Insights and Conditions Survey, whose fortnightly waves record whether exposed firms report price and margin adjustment, workforce reduction, or neither; matching those responses to the exposed SIC divisions over the 2025–26 window would discriminate between the wage-cut and displacement families directly. That exercise is left to further work: it requires a wave-by-wave build the present pipeline does not contain, and we prefer to leave the margin an explicit scenario dimension than to assert a reading of the survey we have not conducted.

4.11 The measured family: the realised outturn as the shock

The validation table invites the obvious next step: discard the elasticity entirely and let the realised outturn *be* the shock. The measured family sets $s_j^{\text{measured}} = \max(\text{fall}_j, 0) \times x_j$—each division's realised export fall times its US-export intensity, with divisions whose exports rose taking a zero shock—so no elasticity, no tariff rate and no pass-through-to-demand assumption enter; the export-side demand response, EPD mitigation included, is read off the data. Each fall is a proportional rate—one minus the ratio of the post-tariff window to the baseline window over the same ten calendar months (May to February) in different years, so seasonality nets out and the quantity is dimensionless rather than a partial-year level—applied as if it persisted over a full year, since the microsimulation is annual, and imposed on the 2026 tax–benefit system and 2026 uprated incomes. Because the outturn embeds the deal, this family has a single tariff scenario rather than a full/EPD pair; it is run under the displacement (20 draws) and wage-cut margins with the machinery of Section 3.3 unchanged, including the maintained assumption that the full export fall passes through to the sector wage bill.

The measured shock is *smaller than either calibrated schedule*: the aggregate gross earnings loss is £0.95 billion a year, against £1.77 billion under the full schedule and £1.39 billion under the EPD. That it lands below even the EPD is the validation table's cross-section translated into pounds: the divisions where the calibrated model over-predicts (aerospace, steel, electrical equipment, computers—realised falls near zero or negative) carry more of the exposed wage bill than the divisions where it under-predicts (apparel, pharmaceuticals), and the clipping of realised export *rises* to zero shocks removes loss the model imputes. Read against the calibrated families, the outturn says the first post-tariff year delivered roughly half the full-schedule earnings shock—the EPD working as priced on autos, but also rebound, re-routing and resilience the one-parameter demand curve does not capture. The implied headline results scale accordingly: displacement puts $16{,}600 \pm 2{,}600$ workers out of work (against 34,200 full, 27,700 EPD), costs the Exchequer £0.31 billion $\pm$ 0.11 a year (wage-cut margin: £0.53 billion, deterministic), and raises relative BHC poverty by 0.028 ± 0.010 percentage points (wage cut: zero).

The structural findings survive the change of shock family intact, which is the point of running it. The cushioning gap reappears with the same sign and nearly the same magnitude: the system absorbs 31.6±5.5 per cent of the measured displacement loss across the 20 draws (29.4 per cent at seed 0) but 41.6 per cent of the earnings-equivalent wage cut—the marginal-versus-average tax arithmetic and the gated UC extensive margin of Section 4.3 operate identically on a differently-shaped sector shock. The Gini moves by $+0.00015$ (displacement) and -0.00007 (wage cut); poverty is again a displacement-margin phenomenon only. The measured family thus separates the paper's two kinds of claim: the pound-denominated *levels* depend on the shock family and are roughly halved relative to the full schedule—readers should treat the measured levels as the current best estimate of the realised first-year shock—while the *orderings* this family can test—wage cuts cushioned more than job losses, and fiscal incidence dwarfing poverty incidence—are properties of the tax–benefit rules that the change of shock family does not overturn. The measured shock was not run under the inactivity or reallocation margins, and no measured regional exhibit is reported, so those orderings are established only for the calibrated families.

5 Policy

5.1 What the Economic Prosperity Deal was worth

Priced as the difference between the two tariff scenarios run through identical machinery (Section 4.7), the EPD is worth, per year on the displacement margin: about 6,500 manufacturing jobs (the mean paired-draw difference over 50 displacement draws, $\pm$4,388, positive in 47 of 50; the analytic quota expectation is 6,857), £0.38 billion of gross earnings, and roughly £0.2 billion to the Exchequer (£0.19 billion $\pm$ 0.38 mean paired-draw difference on the displacement margin, positive in 36 of 50 draws; £0.22 billion, deterministic, on the wage-cut margin). Relative BHC poverty impacts fall from $+0.055 \pm 0.021$ to $+0.042 \pm 0.022$ percentage points under displacement. These are modest absolute numbers—the direct earnings channel of the full schedule was itself only 0.06 per cent of GDP—but they are not evenly spread. The deal's value is a bundle of three sectoral clauses with three distinct geographies. The auto quota (100,000 vehicles at 10 per cent, approximately the 2024 US export volume, so the in-quota rate is the effective one) removes the single largest derived shock and is worth the most to the West Midlands and the wider vehicle supply chain. The pharmaceutical exemption removes the SIC 21 employment shock entirely, protecting a high-wage, geographically dispersed sector—but its price was paid through a different fiscal door (below). Steel relief, modelled at half the 25 per cent rate, is the weakest clause: it was conditional and slow to implement, it leaves Yorkshire and the Humber still exposed under EPD displacement (a Monte Carlo mean loss of £34 $\pm$ 37 per person per year; its worst single draw reaches £184, and £209 under the full schedule), and in any case the steel sector's binding trade-policy exposure is EU/UK safeguards, not the US tariff. What the deal did *not* do is also visible in Table 1: machinery, electronics, aerospace and chemicals—which together account for more derived earnings loss than autos—remain at the 10 per cent baseline, which is why the EPD removes only about a fifth of the gross earnings shock.

5.2 The pharmaceutical channel

The pharmaceutical exemption deserves its own accounting because its incidence is fiscal and price-side, not an employment shock—the cleanest illustration in this paper of statutory versus economic incidence (Atkin et al., 2025). Statutorily, the December 2025 side agreement zero-rates UK pharmaceutical exports to the US for three years. Economically, the exemption was purchased through the NHS: the VPAG rebate rate on branded medicine sales was cut from 22.5 to 14.5 per cent, alongside NHS pricing concessions (ABPI, 2025). With UK branded-medicine sales subject to VPAG in the tens of billions, an eight-percentage-point rebate cut is a large number by the standards of this paper's Exchequer figures. This is explicitly a back-of-envelope order-of-magnitude remark rather than a costing: the arithmetic is (VPAG-eligible branded sales base) $\times$ 0.08, and this paper neither builds that sales base nor models the NHS pricing concessions that travel with it, so no figure is asserted. The qualitative point stands and is the one that matters—the exemption was purchased, not granted, and the price sits on the health budget rather than in household earnings. This paper's employment-channel machinery correctly assigns pharma a zero shock under the EPD, but the reader should not conclude the exemption was free: its cost simply does not pass through household earnings, and therefore does not appear in a tax–benefit microsimulation. A full welfare accounting of the EPD would set the £0.2 billion annual Exchequer saving found here against the VPAG concession on the other side of the public balance sheet.

5.3 Cushioning reforms

The cushioning result of Section 4.2 reframes the standard policy menu. The decomposition of Section 4.3 shows the binding constraint on cushioning displaced workers is not the generosity of out-of-work support but the gates on it: on these data, modelled take-up and the joint partner-income taper leave two-thirds of displaced workers' families with no UC at all, so reforms that raise UC rates or cut the taper do little for the displaced worker whose family never enters an award calculation. Three directions follow, all simulable in this framework. First, gate easements for the involuntarily displaced: automatic enrolment (or an assumed-take-up presumption) after redundancy, easing the joint partner-income assessment, and disregarding savings for a fixed period—the last consistent with, though not testable on, FRS data that carry no household capital—would extend to job losers part of the automatic cushioning that wage-cut workers already enjoy. Even full UC coverage would not close the 33-versus-41 per cent gap by itself, however: most of the gap is income-tax arithmetic (marginal versus average rates), not benefit design. Second, extending the duration or level of contribution-based new-style JSA, the instrument that reaches displaced workers regardless of take-up frictions and partner income—a UK analogue of earnings-related unemployment insurance. Third, wage insurance in the spirit of Hyman et al. (2024), which subsidises re-employment at lower wages: in the language of this paper it converts a displacement-margin realisation of the shock into a wage-cut-margin one, and Section 4.2 shows the UK system is already better at absorbing exactly that margin—so wage insurance and the existing UC taper are complements, not substitutes. Throughout, the revenue-recycling caveat of Ignatenko et al. (2025) applies: this paper prices reforms in a first-round world with fiscal policy otherwise fixed, and financing choices can alter even the sign of aggregate employment effects.

6 Discussion and conclusion

This paper asked who bears the 2025 US tariffs on UK goods after the tax–benefit system responds. Three conclusions stand.

First, the adjustment margin, not the tariff rate alone, determines what the shock *is* when it reaches households—and the direction of that determination reverses the naive prior. The UK tax–benefit system cushions an in-work wage cut more than an earnings-equivalent wave of job losses, because income tax relieves a wage cut at marginal rates but a lost job only at average rates, and Universal Credit—though it responds more to displacement, as intuition predicts—reaches only a third of displaced workers' families, gated by take-up and the joint partner-income taper. Statutory generosity toward the unemployed is not economic generosity once the gates bind on the population actually displaced. The fiscal ordering is the mirror image, though suggestive rather than sharply identified: the wage-cut margin costs the Exchequer more than the mean displacement draw, by a gap that sits inside one draw-to-draw SD—while only the displacement margin moves poverty at all.

Second, on the displacement margin the shock is regionally, not vertically, concentrated. Displaced manufacturing earnings sit in deciles 7–10, so national inequality and poverty statistics barely register a shock that is nonetheless a £100–200-per-head event in whichever handful of regions a given realisation lands on. Trade-shock incidence in the UK is a geography problem wearing the clothes of a distribution problem—consistent with the China-shock literature's regional persistence findings (Autor et al., 2021; Dix-Carneiro and Kovak, 2017) but established here at the household, rules-based level.

Third, the Economic Prosperity Deal was worth about 6,500 jobs and roughly £0.2 billion a year to the Exchequer on the direct channel—real but modest, concentrated in autos, with the pharmaceutical clause's true cost routed through the NHS rather than the labour market and the steel clause largely beside the sector's EU-dominated exposure.

Limitations. Seven are worth stating plainly. (1) The inactivity margin assumes every displaced worker aged 50 and over passes the Work Capability Assessment, which is deliberately an upper bound on the health-element route; no assessment process, waiting period or conditionality regime is modelled, though the WCA is a gatekeeping institution whose pass rates respond to local labour-market slack (Roberts and Taylor, 2022) and the health-related caseload was already at a post-pandemic high when the tariffs arrived (IFS, 2024). Realised flows could sit anywhere between the 50-per-cent sensitivity variant and the upper bound. (2) Monte Carlo dispersion is large relative to the means, so only orderings that survive it are asserted; the displacement–wage-cut fiscal gap is identified by the aggregate cushioning accounting rather than by the draws. (3) Absolute BHC poverty against a fixed baseline line is computed outside the model, which carries no such variable in this version. (4) The central analysis covers the direct exposed-sector earnings channel only. Upstream propagation—which the production-network literature finds can multiply direct effects severalfold (Barrot and Sauvagnat, 2016; Acemoglu et al., 2016)—is bounded in Appendix F, where a single Leontief round roughly doubles the gross earnings loss and leaves every structural ordering intact. Still omitted are downstream and induced-consumption rounds, local-multiplier ripple effects put at 0.4–1.6 additional local jobs per lost tradable job (Moretti, 2010; Gathmann et al., 2020), general-equilibrium wage responses, the exchange-rate offset (Corsetti et al., 2025), and retaliation; fiscal policy is held fixed throughout, and Ignatenko et al. (2025) show revenue recycling can flip aggregate employment signs, so the fiscal numbers are first-round incidence, not equilibrium outcomes. The direct shock is accordingly a lower bound: compounding the upstream round with local multipliers implies an all-channel figure of roughly 0.16–0.31 per cent of GDP, approaching the OBR's one-third to three-quarters of a per cent from below. (5) No behavioural margin is active on the household side. The reallocation family does move workers between sectors, answering the objection that after the China shock displaced manufacturing workers went into healthcare, delivery and retail rather than staying unemployed (Autor et al., 2014; Dauth et al., 2021); but who reallocates is *assumed* rather than estimated, no worker reallocates partially or fails to reallocate, and the sector switch is fiscally inert, so all the economic content sits in the earnings and hours penalty. Added-worker responses, retirement timing, self-employment entry (Irastorza-Fadrique et al., 2025) and labour-supply responses to the tax and benefit changes themselves are all suppressed. (6) The constituency layer is illustrative only: its industry codes are imputed on a different dataset and period, so it cannot reproduce observed local industry mix, and its seat ranking should not be read against exposure-based analyses that can (City-REDI, 2025; Centre for Cities, 2025). (7) The trade numerator is on the HMRC customs basis, which sits below the ONS balance-of-payments basis; intensities inherit that coverage gap.

Further work. Beyond modelling the WCA gate itself, three directions: a measured-margin family calibrated to the accumulating outturn, replacing the stylised poles with the mix of displacement and wage cuts the data actually chose, in the spirit of Rud et al. (2022)'s decomposition; sharpening the constituency geography with observed local industry structure (BRES employment by SIC) in place of the demographic imputation; and dynamic transition paths—re-employment hazards, scarring, and the multi-year fiscal tail that Dix-Carneiro (2014) and Beatty and Fothergill (2017) both imply a one-year snapshot understates.

The bottom line inverts where the paper began. The 2025 tariff reaches UK households mainly as a public-finance event, not a poverty event; and the arm of the welfare state that absorbs it best is not the safety net for the jobless but the automatic machinery that relieves every pound of in-work earnings at the marginal tax rate, with no claim to make and no gate to pass.

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A Data appendix: the trade-by-SIC build

The US-export intensity table x_j (Table 1) is built from public sources, with all raw API pulls cached alongside the replication materials.

Numerator. UK goods exports to the United States, calendar 2024, in pounds, from the HMRC `uktradeinfo` OData API—the machine-readable publication of HMRC Overseas Trade Statistics (OTS) and Regional Trade Statistics (RTS). The RTS endpoint is queried for CountryId 400 (United States), non-EU export flows, months 202401–202412, summed over all Government Office Regions, keyed by 2-digit SITC (rev. 4) chapter. Nine chapters that straddle SIC divisions (SITC 33, 66, 68, 71, 77, 78, 79, 87, 89) are refetched from the OTS endpoint at 3-digit SITC and split—for example, 714 aero engines map to SIC 30 (other transport) rather than SIC 28 (machinery)—with the build asserting that each OTS split reproduces its RTS chapter total to within rounding.

Crosswalk. SITC (rev. 4) to SIC 2007 division, coded explicitly in the build script. It is the standard manufacturing concordance plus documented judgement calls: SITC 971 (non-monetary gold), 681 (silver and platinum group) and 896 (works of art) are excluded as London bullion- and art-market re-exports rather than UK production employment—the two precious-metal exclusions are applied identically in the 2024 intensity build and in the monthly outturn build, and are quantitatively decisive rather than cosmetic, removing £24.2 billion of gross 2024 flow between them (£16.3 billion of non-monetary gold, £7.9 billion of silver and platinum group); in January 2025 non-monetary gold alone was £6.1 billion against a non-gold monthly total of roughly £4 billion, so a build retaining it would track bullion arbitrage rather than traded manufactures, and this is the same exclusion ONS applies to its published monthly series, which keeps our falls and theirs on a comparable commodity base; 891 arms and ammunition maps to SIC 25; 872 medical instruments to SIC 32 with 871/873/874 to SIC 26; 776 semiconductors to SIC 26 with the rest of chapter 77 to SIC 27; 333 crude oil is dropped (extraction, negligible FRS employment) while 334/335 refined products map to SIC 19. Excluded chapters (91, 93, 96, 97 and non-goods special transactions) total under £0.3 billion.

Validation. The 2024 all-commodity total on the RTS/OTS customs basis is £55.6 billion, of which £52.5 billion maps to SIC divisions after the documented exclusions; the ONS balance-of-payments basis is about £59.3 billion, the difference being BoP coverage and valuation adjustments. Sector cross-checks pass: SITC 78 road vehicles £9.0 billion against SMMT's £8–9 billion; SITC 54 medicinal and pharmaceutical products £6.0 billion against ABPI's £6.6 billion (BoP basis). The US share of UK goods exports implied by these totals (roughly a fifth) matches the survey-based figure in CEP (2025).

Denominator. Division total turnover from the ONS Annual Business Survey (non-financial business economy, Section C sheet), taken at each division's latest non-suppressed year (2023 for most divisions; beverages fall back to 2022 owing to confidentiality suppression). Turnover proxies gross output at basic prices; the intensity is US exports over turnover. Divisions with no US exports or suppressed turnover are omitted and take a zero shock. Note the vintage mismatch in the measured family: the denominator is 2023 ABS turnover (the latest non-suppressed year), while the export falls it scales are the realised 2025–26 outturn. Nominal turnover growth between 2023 and 2025–26 therefore makes the denominator slightly too small and the measured intensities—and hence the measured shocks—marginally too large; no reflation is applied, so the measured family is conservative in the direction of a larger shock.

The measured outturn build. The realised export falls behind the measured family (Sections 4.10–4.11) are built from a monthly pull of the same `uktradeinfo` source: UK goods exports to the United States, January 2023 to February 2026, from the OTS endpoint grouped server-side by month and SITC code (the RTS endpoint, used for the 2024 intensity build, stops providing complete monthly coverage after April 2025), mapped to SIC divisions with the crosswalk above; the pull reproduces the 2024 calendar total of the intensity build (£52.5 billion) exactly. The post-tariff window is May 2025 to February 2026. April 2025 is excluded on two grounds: it mixes the initial collapse with the unwinding of the Q1-2025 front-running spike, and it is the month the elasticity was calibrated on, so excluding it keeps the validation out of sample. The baseline for each post month is the *mean of the same calendar month one and two years earlier*: comparing like calendar months nets out seasonality, and averaging over two years dilutes the tariff-anticipation contamination of the one-year base—exports were pulled forward into Q4-2024/Q1-2025, so a pure year-on-year comparison for the January–February 2026 post months would divide by an inflated base and overstate the fall. The choice turns out to matter little in aggregate (a 12.7 per cent fall on the two-year-average baseline against 12.5 per cent on pure year-on-year) but more for individual divisions with concentrated front-running, notably pharmaceuticals (30.7 against 22.5 per cent); the one-year variant is stored alongside as a sensitivity column. Pharmaceuticals is the division where this choice matters most and is best reported as a span rather than a point: on a calendar-2025-versus-2024 basis the same data give 10.9 per cent against 30.7 on the two-year-average baseline, with the ONS's published 16.2 per cent inside the range. The crosswalk is exact for this division—SIC 21 is SITC 54, the ONS's own "medicinal and pharmaceutical products" aggregate—so the divergence is a baseline effect and not a mapping error. Motor vehicles behave the same way on a smaller scale (27.4 per cent on the calendar baseline against 32.0 on the two-year, ONS 24.4), and in both cases the two-year baseline's deliberate dilution of the front-run base is the main source of the gap to the published figures. Divisions whose US exports rose over the window take a zero measured shock—the displacement machinery cannot model employment gains—with the raw negative falls preserved in the packaged division-level output. It is worth being explicit about what the resulting object is. Each division's realised fall is a *proportional rate*, one minus the ratio of the post-tariff window to the baseline window, and both windows cover the same ten calendar months (May to February) in different years, so seasonality nets out and the quantity is dimensionless. It is not a partial-year level and is never treated as one. Two assumptions follow. First, because the microsimulation is annual, the ten-month average proportional fall is applied as though it persisted over a full year. Second, the shock is measured over 2025–26 but imposed on the 2026 tax–benefit system and 2026 uprated incomes, so the counterfactual asks what the measured shock would do to households under 2026 rules rather than reconstructing the calendar of the outturn itself.

Reconciliation with the ONS published series. Our aggregate 12.7 per cent fall and the widely quoted ONS figure of 24.7 per cent are not competing estimates of the same quantity, and it is worth stating the difference precisely. The ONS number is an April 2025 *month-on-month* fall against the front-run March 2025 peak, on a series that already excludes non-monetary gold; ours is a ten-month window (May 2025 to February 2026) against a two-year same-calendar-month baseline. Computed on ONS's own basis, our data give a 21.7 per cent April month-on-month fall (£4.85 billion to £3.80 billion) against ONS's 24.7 per cent. The residual three points are the customs-versus-balance-of-payments basis difference documented above—our 2024 mapped total is £52.5 billion of a £55.6 billion all-commodity customs total, against an ONS balance-of-payments figure of about £59.3 billion—and not a discrepancy in the underlying movement. Two further checks. Had precious metals been retained, the same April calculation would read a 44.2 per cent fall, which is precisely the bullion-driven distortion the ONS exclusion exists to avoid and confirms that the exclusion is doing real work rather than trimming a rounding error. And every post-window month lies in the £3.34–4.22 billion range against a 2024 monthly range of £4.07–4.83 billion, so every post month sits below every 2024 equivalent: the sustained-below-baseline reading of the window does not depend on the choice of baseline construction.

B Shock mechanics

Displacement draw. For each exposed division j, the head-count quota is $s_j \times W_j$, where W_j is the division's weighted employee count. Division members are ordered by uniform random keys—so a represented worker's inclusion probability does not depend on their record's grossing weight—and survey weights are consumed against the quota in that order. The quota-crossing record is included with probability equal to the remaining quota fraction of its weight, so the expected displaced weight equals the quota exactly. Displaced workers move to employment status UNEMPLOYED with employment income, hours, employee and salary-sacrifice pension contributions, and statutory maternity, paternity and sick pay all zeroed—without which a displaced worker would remain classified in-work (retaining, for example, childcare entitlements) or keep deducting pension contributions from zero earnings. The pipeline verifies that every displaced person's employment status actually changed in the shocked simulation and raises a hard error otherwise: a silently rejected input would alter entitlements in every downstream result.

Wage-cut earnings-equivalence. The displacement quota removes fraction s_j of division j's weighted head count with ordering keys independent of earnings, so the expected earnings removed is $s_j B_j$, with B_j the division wage bill, up to within-draw selection covariance. The wage-cut family cuts every employee in division j by exactly s_j, removing $\sum_j s_j B_j$ identically—the same expected aggregate loss, delivered on the intensive margin, with conservation asserted at run time rather than assumed. The margin comparison therefore holds the aggregate gross earnings loss fixed by construction.

Inactivity margin. The displacement draw, with displaced workers aged 50 and over (the threshold is a scenario parameter) transitioned to OTHER_INACTIVE rather than UNEMPLOYED, verified with the same hard-error contract, and flagged as having limited capability for work-related activity so that their Universal Credit includes the LCWRA health element. Implementation detail: in policyengine-uk 2.89.2 the benunit-level `uc_LCWRA_element` is a *stored input* of the FRS dataset (imputed survey receipt), so setting the person-level limited-capability-for-work-related-activity flag alone never reaches computed UC—the stored array shadows the formula. The pipeline therefore overrides the element directly: baseline stored element plus one annual LCWRA amount (£217.26 × 12) per benefit unit containing any newly flagged member, with all other households' stored elements untouched; recomputing the element from the model formula instead would reprice the whole population off `is_disabled_for_benefits` and break comparability with the baseline simulation. Because Universal Credit pays at most one health element per benefit unit, the add-on is a benefit-unit indicator rather than a count of flagged members: two flagged members of the same benefit unit (a displaced over-50 couple) receive one element, not two. A hard error is raised if any flagged person's benefit unit lacks the element, or if any benefit unit's add-on exceeds a single annual element, in the shocked simulation. (In the reported runs no benefit unit ever contains two flagged members at these seeds, so the distinction leaves every reported number unchanged.) The `lcwra_takeup` scenario parameter thins the flag within the inactive set (drawn from a separate random stream, so the displacement draw is unchanged); the default 1.0 is the paper's upper-bound assumption and 0.5 the sensitivity variant. At seed 0 under the full schedule, 33 per cent of the displaced are 50 or over, so the margin's scope is roughly a third of the displaced population.

Reallocation margin. The same displacement draw on the same seed—the two families are paired worker-for-worker—but drawn workers are re-employed instead of transitioned to unemployment. Destinations are SIC 47 (retail), 86 (human health), 49 (land transport, including delivery and couriers) and 56 (food and beverage service), drawn per worker in proportion to those divisions' FRS employee grossing-weight shares within the four (0.2951, 0.4581, 0.0920, 0.1548), and written to `sic_industry_division`. Annual earnings are scaled by $(1-p)(1-\ell/12)$, where p is the earnings penalty (0.283 central, 0.140 in the hours/age-controlled low-penalty variant) and ℓ the months out of work before the services job starts (0 central, 3 in the lag variant); annual hours are scaled by the lag factor alone. The scenario constructor hard-errors if $p \notin [0, 1)$ or $\ell \notin [0, 12]$, and the same run-time contract as the other draw-based families verifies that the intended transition actually landed in the shocked simulation rather than failing silently. Two properties of the design should be kept in view when reading the table: the family removes only the penalty on the movers, so its gross loss (0.283× the displacement gross loss under the central penalty) is far below the earnings-equivalent aggregate the other families remove; and `sic_industry_division` is a pure input in policyengine-uk 2.89.2 with no downstream consumer in the tax–benefit calculation, so the destination is reportable but fiscally inert—the entire economic content is the earnings and hours penalty.

C Monte Carlo details and manual measures

Draws and seeds. Each displacement scenario is run for 50 draws with seeds $0, \ldots, 49$; reported values are means with sample standard deviations ($n-1$). Per-draw decile and regional profiles are stored for every draw, and the decile, regional and age-band numbers in the text are Monte Carlo means with SDs across the 50 draws. The wage-cut margin is deterministic and collapses to a single run; the inactivity margin is recomputed with its own runs over 20 draws (seeds $0, \ldots, 19$, shared with the displacement seeds so inactivity–displacement differences are paired and largely difference out the draw noise; Appendix B). Draw-to-draw dispersion is substantial—the Exchequer-cost SD is about £0.30 billion against means of £0.7–0.9 billion—because with roughly 30,000 displaced workers nationally, single high-weight, high-earnings FRS records move aggregate outcomes; the text applies the discipline of narrating only orderings that survive the SD. Full and EPD displacement scenarios share seeds, so paired-draw differences partially difference out the draw noise (the full−EPD Exchequer difference is £0.19 billion ± 0.38 and positive in 36 of 50 paired draws; the deterministic wage-cut Exchequer cost exceeds the displacement draw's in 37 of 50 draws under each schedule).

Absolute poverty with a fixed line. The installed model version exposes relative BHC and AHC poverty but no absolute-poverty variable. Absolute poverty changes are therefore computed manually: the baseline BHC poverty line (60 per cent of the *baseline* equivalised median) is held fixed, and the shocked population is evaluated against it, so the measure captures absolute impoverishment rather than movement of the line itself. At seed 0 the fixed-line changes are +0.058 percentage points (full displacement), +0.029 (EPD displacement) and zero on the wage-cut margin under either schedule (the inactivity margin's poverty changes differ from displacement's only marginally, and never upward; Section 4.4)—close to the relative-line changes, confirming the shock is too small to move the national median materially.

Cushioning accounting. The cushioning rate of Section 4.2 is $1 - \Delta Y/\Delta E$, where ΔE is the weighted gross employment-earnings loss and ΔY the weighted aggregate fall in HBAI household disposable income, both at seed 0. Because it is a ratio taken within a draw, draw noise in the numerator and denominator largely cancels—which record is drawn moves gross and net losses together—so the 33-versus-41 per cent displacement/wage-cut comparison is far less draw-sensitive than the Exchequer-cost levels; the same ordering holds under both tariff scenarios (32.8 vs 41.0 full; 35.1 vs 40.8 EPD).

D Sensitivity grid: elasticity and pass-through

The derived shock (Equation 1) depends on the elasticity ε and the pass-through π only through their product $\varepsilon\pi$, so the grid in Table 10—$\varepsilon \in \{1.0, 1.5, 2.0, 3.0\} \times \pi \in \{0.5, 0.75, 1.0\}$, full schedule, displacement (10 draws per cell, seeds 0–9) and the deterministic wage cut—has a predicted structure, which the computed cells confirm. First, *levels are linear in* $\varepsilon\pi$: the wage-cut gross earnings loss per unit of $\varepsilon\pi$ is £0.886 billion in every cell to machine precision (no cell's sector shock reaches the clip at 100 per cent of the wage bill), and cells with equal products are identical—for example $(\varepsilon, \pi) = (2.0, 0.5)$ reproduces $(1.0, 1.0)$ exactly. The displacement gross loss per unit of $\varepsilon\pi$ varies only within 13 per cent across cells, which is quota rounding and draw noise, not curvature. Second, *rates are invariant*: the displacement cushioning rate stays within 32.9–33.9 per cent and the wage-cut rate within 40.6–41.5 per cent across a six-fold range of shock sizes, and the wage-cut rate exceeds the displacement rate in all twelve cells. The paper's structural claims—the cushioning gap, its decomposition, and every ordering—are therefore properties of the tax–benefit rules, not of the calibrated $\varepsilon = 2.0$; only the pound-denominated levels (and the near-proportional poverty effect) move with $\varepsilon\pi$, and they move linearly, so readers preferring a different anchor can rescale directly.

E Duration and take-up sensitivities

Table 11 reports two sensitivities on the full-schedule displacement margin, each over the same 10 draws (seeds 0–9) so columns are directly comparable to the recomputed central case.

Table 10: Sensitivity grid over $\varepsilon \times \pi$, full tariff schedule

			Displacement (10 draws, mean ± SD)					Wage cut	
ε	π	$\varepsilon\pi$	Gross loss (£bn)	Exchequer (£bn)	Cushion (%)	Displaced (000s)	Poverty (pp)	Gross loss (£bn)	Cushion (%)
1.0	0.50	0.50	0.40	0.16 ± 0.10	32.9 ± 5.2	7.9	+0.014	0.44	41.0
1.0	0.75	0.75	0.69	0.29 ± 0.09	33.9 ± 4.4	13.2	+0.022	0.66	41.5
1.0	1.00	1.00	0.87	0.37 ± 0.11	33.9 ± 3.5	17.2	+0.030	0.89	41.3
1.5	0.50	0.75	0.69	0.29 ± 0.09	33.9 ± 4.4	13.2	+0.022	0.66	41.5
1.5	0.75	1.13	0.94	0.40 ± 0.11	33.9 ± 3.3	18.6	+0.031	1.00	41.3
1.5	1.00	1.50	1.32	0.55 ± 0.14	33.0 ± 2.8	26.5	+0.044	1.33	41.3
2.0	0.50	1.00	0.87	0.37 ± 0.11	33.9 ± 3.5	17.2	+0.030	0.89	41.3
2.0	0.75	1.50	1.32	0.55 ± 0.14	33.0 ± 2.8	26.5	+0.044	1.33	41.3
2.0	**1.00**	**2.00**	1.68	0.70 ± 0.16	32.9 ± 2.6	32.9	+0.056	1.77	41.0
3.0	0.50	1.50	1.32	0.55 ± 0.14	33.0 ± 2.8	26.5	+0.044	1.33	41.3
3.0	0.75	2.25	2.04	0.87 ± 0.23	33.5 ± 4.6	40.4	+0.063	1.99	40.9
3.0	1.00	3.00	2.62	1.10 ± 0.26	33.1 ± 4.3	52.1	+0.084	2.66	40.6

Notes: full tariff schedule, period 2026. Displacement cells are means ± SD over 10 Monte Carlo draws (seeds 0–9); the central calibration ($\varepsilon = 2.0$, $\pi = 1.0$, bold) differs slightly from the 50-draw headline numbers because of the smaller draw count. The wage-cut margin is deterministic; its small cushioning-rate variation (40.6–41.5 per cent) reflects the sector composition of the marginal shock, not $\varepsilon\pi$. Cells with equal $\varepsilon\pi$ are identical by construction. Poverty is the relative BHC rate change in percentage points. Displaced counts are weighted workers in thousands.

Out-of-work duration. The central results annualise a twelve-month spell. Under a six-month expected duration, displaced workers keep 50 per cent of baseline annual earnings, evaluated *in-model* so taxes and means-tested benefits respond to the actual annual income (a documented hybrid, following the sister study's convention: the annual model has no intra-year timing, so the displacement transition—hours zeroed, status unemployed—is unchanged while half a year's earnings are retained). The gross loss halves by construction; the annualised Exchequer cost falls roughly pro rata, in fact slightly more than half (£0.70 to £0.41 billion, 59 per cent of the full-year figure) because the retained earnings slice is taxed above the displaced worker's average rate; the cushioning rate rises from 33 to 42 per cent for the same marginal-versus-average-rate reason that drives the paper's headline gap; and the poverty effect nearly vanishes—half a year of earnings keeps most displaced households above the line.

UC take-up. PolicyEngine models UC take-up stochastically through a benunit-level take-up indicator in the dataset (the mechanism decomposition of Section 4.3 found 55 per cent of zero-UC displaced workers are in families modelled as not taking up). Forcing 100 per cent take-up in *both* the baseline and the shocked simulation—so the deltas are shock effects under full take-up, not take-up effects—more than doubles the UC response (£0.16 versus £0.07 billion) and raises the displacement cushioning rate to 38.6 ± 2.5 per cent, closing about seventy per cent of the displacement–wage-cut gap. Modelled non-take-up is thus the single largest reason UC fails to reach the displaced; the remainder is chiefly the joint partner-income taper.

Table 11: Duration and take-up sensitivities, full-tariff displacement (10 draws)

	Central (12-month, modelled take-up)	Duration 6 months	Take-up 100 per cent
Gross earnings loss (£bn/yr)	1.68 ± 0.32	0.84 ± 0.16	1.68 ± 0.32
Exchequer cost (£bn/yr)	0.70 ± 0.16	0.41 ± 0.09	0.79 ± 0.16
Cushioning rate (%)	32.9 ± 2.6	42.2 ± 3.3	38.6 ± 2.5
UC response (£bn/yr)	0.07 ± 0.03	0.01 ± 0.01	0.16 ± 0.06
Poverty change, BHC (pp)	$+0.056 \pm 0.014$	$+0.013 \pm 0.010$	$+0.048 \pm 0.019$

Notes: means ± SD over 10 Monte Carlo draws (seeds 0–9), shared across columns. The take-up column compares a forced-take-up shocked simulation against a forced-take-up baseline (person-weighted modelled take-up in the dataset: 55 per cent). The duration column's cushioning rate is computed against its own halved gross loss.

F Supply-chain amplification

The core design covers the direct exposed-sector channel only. This appendix bounds the first omitted channel—upstream propagation through the domestic production network, which the firm-level literature finds can multiply direct trade-shock effects severalfold (Barrot and Sauvagnat, 2016; Acemoglu et al., 2016)—with a Leontief input-output round, and reruns the full-tariff scenario on the amplified shock.

Method. Let f be the vector of direct export-demand falls in gross-output terms, $f_j = \varepsilon\,\tau_j \times$ (US exports of division j)—the same primitives as Equation (1), in pounds rather than as a wage-bill share, using the identical HMRC export numerator as the intensity build (Appendix A). With A the domestic technical-coefficient matrix from the ONS UK input-output analytical tables (product-by-product, 2022 revised edition, 104 CPA products, domestic use at basic prices) and $L = (I - A)^{-1}$ the Leontief inverse, the upstream requirement falls are

$$\Delta g \;=\; (L - I)\, f, \tag{2}$$

where subtracting the identity removes the direct round, which is already the paper's central channel. Each supplying product k converts its output fall to an earnings loss via its compensation-of-employees share of gross output from the same tables, $\Delta g_k \times (\mathrm{CoE}_k/\mathrm{GO}_k)$, and to an upstream earnings-shock *rate* $s_k^{\mathrm{up}} = \pi\,\Delta g_k/\mathrm{GO}_k$, comparable to s_j. CPA products align with SIC 2007 divisions at the two-digit level; products splitting a division are aggregated on gross-output weights, and the few groups spanning divisions (e.g. CPA F41–43) assign each member division the group *rate* (the uniform-rate assumption within the group). Rates and levels are treated differently for these spanning groups: because the rate is a ratio it can be replicated across member divisions, but the earnings-loss and output-fall *levels* are additive and are therefore split across member divisions—on each member's share of the gross output of the products it holds exclusively, equally where no member has one—so that the per-division losses in Table 12 sum to the economy-wide upstream aggregate rather than over-counting the spanning groups. The aggregate loss and the amplification factor are computed product-by-product and so are invariant to this allocation. The person-level scenario then runs on $s_j^{\mathrm{total}} = s_j^{\mathrm{direct}} + s_j^{\mathrm{up}}$ through the unchanged margin machinery.

Amplification. In IO-accounts terms, valuing both channels with the same earnings conversion, the full-tariff direct channel is £2.62 billion of compensation of employees and the upstream round adds £2.50 billion—an amplification factor of **1.95**. This factor is Type-I-equivalent in national-accounts terms—direct plus indirect upstream requirements, with no induced-consumption round—so it may be compounded with local-multiplier estimates without double-counting the induced channel. ONS publishes Type I output multipliers for manufacturing of roughly 1.4–2.0; 1.95 sits at the top of that range, as expected for the input-intensive exposed manufacturing divisions rather than manufacturing as a whole. The FRS realisation reproduces it: the aggregate gross earnings loss under s^{total} is £3.45 billion a year against the direct-only £1.77 billion. A doubling sits comfortably inside the "severalfold" range of the production-network literature, and remains a lower bound on the all-channel effect: the Leontief round adds upstream input suppliers only, with no downstream users, no induced-consumption round and no local multipliers (Section 6).

Who gains exposure. The upstream round pulls services into the exposed set for the first time. Table 12 lists the largest upstream suppliers by earnings loss: wholesale trade alone takes £0.39 billion, followed by motor-vehicle trade, IT services, retail, finance, legal and accounting, employment services, land transport and warehousing—the classic B2B services complex around manufacturing. Within manufacturing, basic metals gains the most exposure in rate terms: its upstream rate of 2.1 per cent of the wage bill nearly doubles its direct 2.5 per cent, as it supplies the tariffed automotive and machinery chains.

Table 12: Largest upstream supplying divisions, full tariff schedule

SIC	Division	Upstream earnings loss (£m/yr)	Upstream shock rate (% of wage bill)
46	Wholesale trade	392	0.57
45	Wholesale and retail trade of motor vehicles	191	0.79
62	Computer programming and consultancy	134	0.25
47	Retail trade	130	0.21
64	Financial services	119	0.26
69	Legal and accounting	119	0.34
25	Fabricated metal products	117	1.08
78	Employment services	105	0.40
49	Land transport	104	0.42
52	Warehousing and transport support	104	0.62

Notes: upstream earnings loss $\Delta g_k \times \text{CoE}_k/\text{GO}_k$ and rate $\Delta g_k/\text{GO}_k$ aggregated to SIC divisions, full tariff schedule, $\varepsilon = 2.0$, $\pi = 1.0$. ONS input-output analytical tables 2022 (revised); HMRC 2024 US export values.

Headline results. Rerunning full-tariff displacement (10 draws, seeds 0–9) and the deterministic wage cut on s^{total}, against the direct-only results over the same seeds: displaced workers rise from 32.9 to 71.4 thousand; the Exchequer cost from £0.70 $\pm$ 0.16 to £1.51 $\pm$ 0.29 billion a year (wage cut: £0.98 to £1.90 billion); the BHC poverty-rate change from $+0.056 \pm 0.014$ to $+0.087 \pm 0.029$ percentage points—sublinear in the doubled gross loss, because upstream displacement is spread thinly across well-paid service divisions—and the wage-cut poverty change becomes slightly positive (+0.031 points) where the direct-only wage cut moved no one across the line. The structural findings survive intact: at seed 0 the displacement cushioning rate is 33.5 per cent against the wage cut's 40.6 (direct-only: 32.8 and 41.0)—the intensive-extensive ordering is a property of the tax–benefit rules, not of which sectors are shocked—and the Gini change stays two orders of magnitude below annual sampling variation (+0.0004 displacement, −0.0002 wage cut). Notably, adding the service divisions does *not* flatten the top-half concentration of Section 4: deciles 7–10 take 75 per cent of the mean displacement income loss under s^{total} against 69 per cent direct-only over the same draws, because the newly exposed wholesale, professional, IT and finance earnings sit at least as high in the distribution as the manufacturing earnings they supply. The regional-not-vertical reading of the incidence is therefore robust to—indeed sharpened by—the first round of supply-chain propagation.

G Geographic incidence

This appendix carries the paper's geography in two layers. The first, below, is the ITL1 or government office region level, which rests on the FRS's own recorded region—an observed variable—and is the geography the main text relies on. The second, in Section G.2, projects the same runs onto Westminster constituencies using an *imputed* industry code, and is illustrative only.

G.1 Regional incidence

Table 13 reports the full three-scenario regional profile; Figure 4 maps its two displacement columns at ITL1/government office region level.

Table 13: Mean fall in household disposable income per person per year by region (£, Monte Carlo means ± SD over 50 draws)

	Full tariffs		EPD
Region	Displacement	Wage cut	Displacement
Wales	108 ± 234	65	75 ± 181
East of England	74 ± 76	61	51 ± 59
East Midlands	72 ± 73	51	70 ± 93
West Midlands	64 ± 56	55	40 ± 54
Scotland	62 ± 50	39	45 ± 37
Northern Ireland	61 ± 46	49	63 ± 52
Yorkshire and the Humber	60 ± 48	47	34 ± 37
London	48 ± 53	52	46 ± 49
South East	47 ± 40	42	41 ± 37
North West	32 ± 34	27	10 ± 13
South West	31 ± 31	36	31 ± 28
North East	22 ± 47	18	23 ± 46

Notes: absolute values of the per-person disposable income change; displacement columns are Monte Carlo means ± draw-to-draw SD over 50 draws, the wage-cut column is the deterministic run. SDs are comparable to or larger than the means—no pairwise regional ordering survives one SD; the Wales mean is dominated by a single high-weight FRS record (Appendix C).

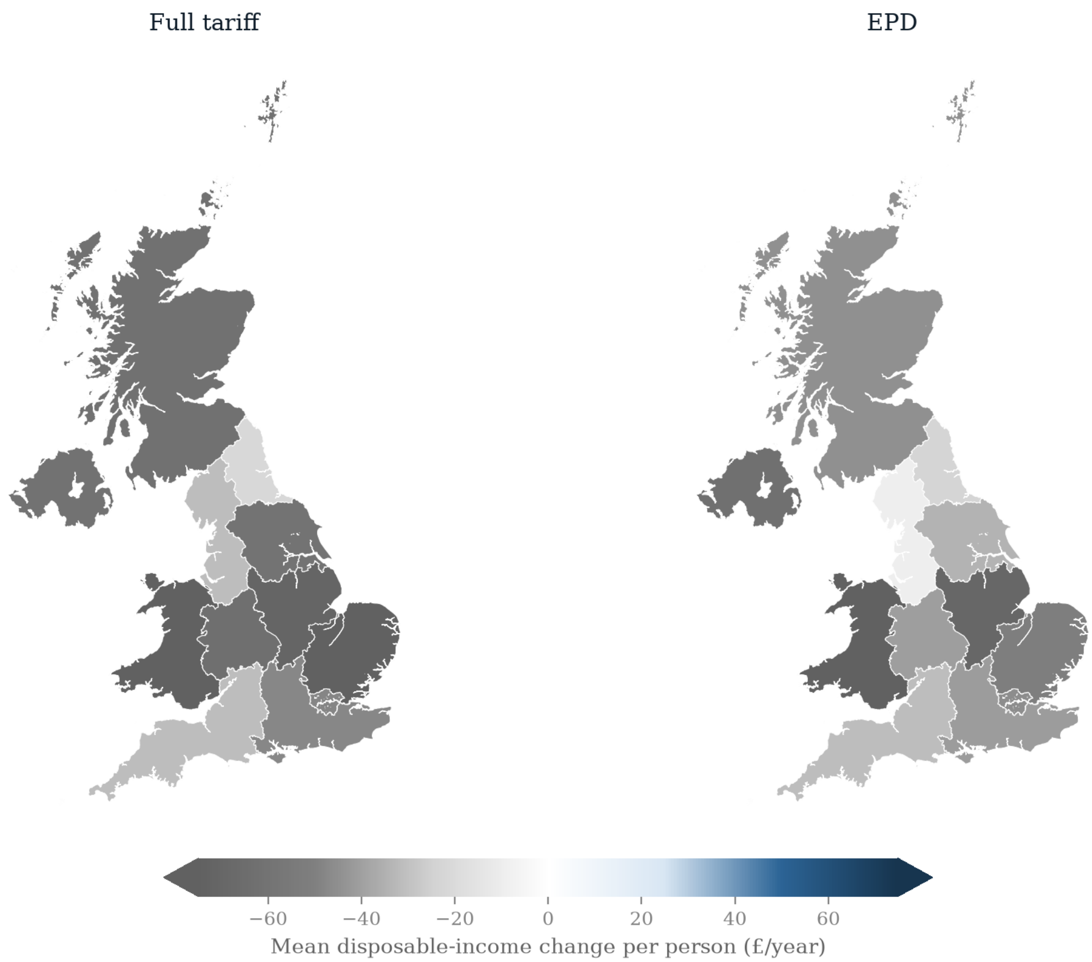

Figure 4: Regional choropleth maps of the mean disposable-income change per person per year (£, Monte Carlo mean over 50 displacement draws), by ITL1/government office region: full-tariff displacement (left) and EPD displacement (right), on a shared diverging scale (losses in grey). The diverging scale is clipped at the 95th percentile of the absolute change (caps show clipping). Region-level draw-to-draw SDs are comparable to the means (Table 13), so the map shows the average geography, not a sharply identified ranking.

G.2 Constituency-level illustration

The constituency layer is deliberately *not* part of the paper's main-text geography. It rests on an industry code that is *imputed*, on a different dataset and a different period, and it cannot reproduce observed local industry mix. That limitation is not cosmetic. Our seed-0 constituency ranking is led by Scottish and Belfast seats, whereas the exposure-based institutional analyses point at the West Midlands (City-REDI, 2025; Centre for Cities, 2025)—and the difference is a direct consequence of the imputation: a demographic-cell imputation reproduces where people of a given age, sex, region and earnings decile live, not where the car plants are. The exhibit is therefore illustrative of the *mechanism* by which a displacement draw concentrates geographically, not an estimate of which seats are hardest hit. Replacing the demographic imputation with observed local industry structure (BRES employment by SIC) is the fix, and is listed under further work (Section 6).

The regional concentration sharpens further at constituency level. Projecting the two displacement scenarios onto the 650 Westminster parliamentary constituencies with PolicyEngine's calibrated $650 \times 53{,}508$ constituency weight matrix (Figure 5) shows the full-schedule loss concentrating in Scottish and Northern Irish seats. These constituency results are a single-draw (seed-0), imputation-based illustration—unlike the Monte Carlo regional means of Section 4.8, the rankings would reshuffle under other draws: in that draw the ten hardest-hit constituencies under full-tariff displacement are West Dunbartonshire ($-$£390 per person per year), Glasgow South West ($-$£362), Dumfries and Galloway ($-$£322), Belfast East ($-$£301), Lothian East ($-$£283), Belfast South and Mid Down ($-$£239), North East Fife ($-$£233), Paisley and Renfrewshire South ($-$£229), Caithness, Sutherland and Easter Ross ($-$£227) and Paisley and Renfrewshire North ($-$£223)—eight Scottish seats, consistent with the beverages division's outsize US-export intensity (Scotch whisky), plus the two most manufacturing-intensive Belfast seats. Under the EPD (same seed-0 draw) the map empties across Great Britain but Northern Ireland's residual exposure dominates: nine of the ten hardest-hit seats are Northern Irish, led by Belfast East ($-$£498), Belfast South and Mid Down ($-$£351) and Belfast West ($-$£302), with Sheffield Central ($-$£103)—a steel seat under the deal's half-relief—the only Great British entrant. West Midlands automotive seats (Solihull West and Shirley, Meriden and Solihull East) sit in the hardest-hit quartile under the full schedule but nowhere near the top ten, because the EPD-relieved auto rate and the discrete displacement draw both dilute their exposure. One methodological caveat governs these numbers, mirroring the sister study's constituency analysis: the weight matrix indexes the *enhanced* FRS 2023–24 dataset, which carries no industry codes, so SIC division is imputed from the plain-FRS 2024–25 weighted distribution within age-band $\times$ gender $\times$ region $\times$ earnings-decile cells (99.4 per cent weighted coverage from the full four-way cell, seed 0). Constituency variation therefore reflects demographic and earnings composition under a single seed-0 draw and model-calibrated weights, not observed local industry mix, and is computed on a different dataset and period (2025) from the headline runs—indicative of relative, not absolute, local impacts.

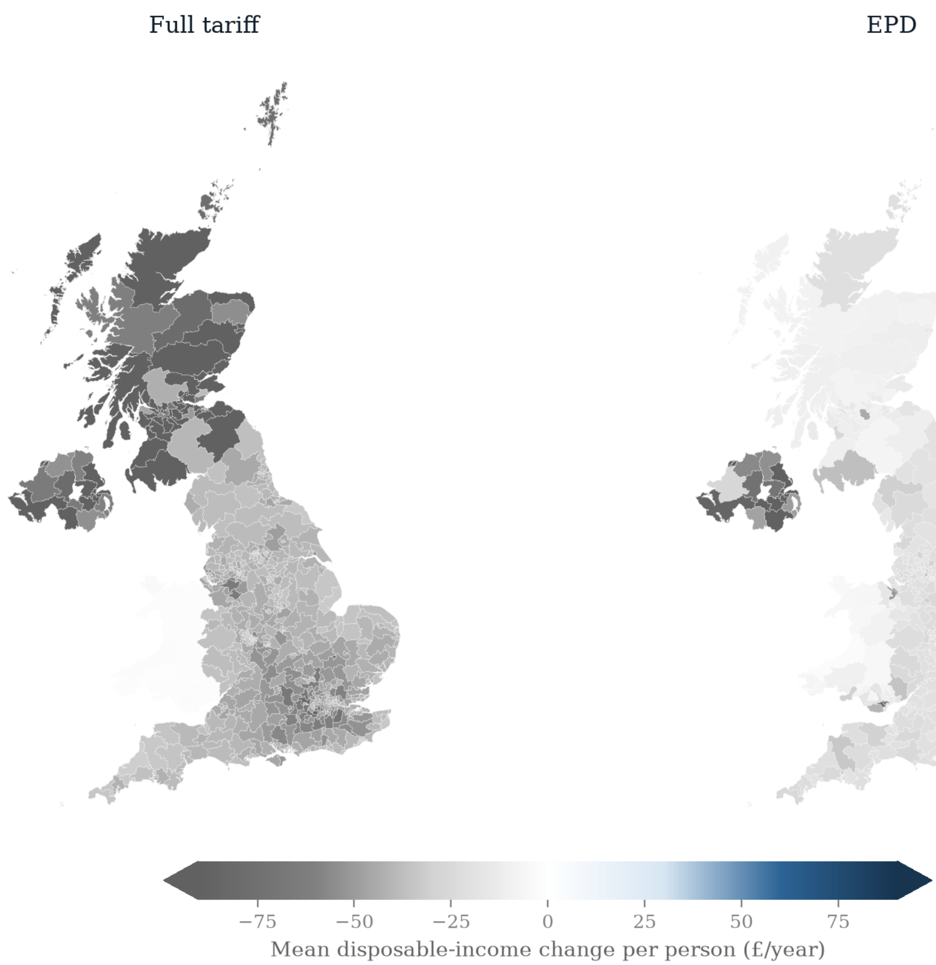

Figure 5: Constituency choropleth maps of the mean disposable-income change per person per year (£, seed-0 draw) across the 650 Westminster parliamentary constituencies: full-tariff displacement (left) and EPD displacement (right), on a shared diverging scale (losses in grey), clipped at the 95th percentile of the absolute change (caps show clipping). Computed on the enhanced FRS 2023–24 dataset with PolicyEngine's calibrated constituency weight matrix and imputed SIC divisions, period 2025; constituency variation reflects demographic and earnings composition under a single displacement draw, not observed local industry mix, and is indicative of relative rather than absolute local impacts.